
Optimal Design for Multinomial Logit Model with Applications to Best Assortment Identification

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Abstract

We study optimal experimental design for multinomial logit (MNL) bandits, where an agent repeatedly selects a subset of K items from a ground set of size N and observes single-choice feedback. Unlike linear or generalized linear bandits, MNL bandits have a *combinatorial* action space, which makes classical optimal design approaches and naive optimization over all subsets computationally intractable. We propose a computationally efficient optimal design framework for MNL models that achieves both statistical efficiency and scalability through two complementary approaches: (i) an exact or certified-approximate reformulation of the design oracle as a 0–1 mixed-integer linear program (MILP) with solver-certified early stopping, and (ii) a fully polynomial-time lifted design that replaces the nonlinear objective with a tractable surrogate. Using the Kiefer–Wolfowitz equivalence theorem, we establish near G-optimality guarantees and characterize the induced statistical–computational trade-offs. As an application, we develop a best assortment identification algorithm for MNL bandits with linear utilities and non-uniform revenues, and prove an instance-dependent sample complexity of $\tilde{O}\left(\frac{d \log N}{\Delta^2}\right)$, where d is the feature dimension, N is the number of arms, and Δ is the minimum revenue gap.

1. Introduction

We study optimal experimental design in multinomial logit (MNL) models (McFadden, 1977) with linear utilities. The MNL model is a fundamental discrete choice model widely used in economics, marketing, and machine learning, where outcomes are generated from subset-dependent multinomial

distributions parameterized by latent utilities. For instance, in online advertising, a user selects (e.g., clicks) one item from a subset of K ads presented on a page. Each ad can be represented by high-dimensional features, such as image or text embeddings produced by an off-the-shelf neural network. This structured feedback setting naturally arises in large-scale applications, where the total number of available items N can be very large.

From an experimental design perspective, this setting poses significant challenges. Unlike linear or generalized linear models—where the design space consists of individual arms—the design space in MNL models is *combinatorial*: each experiment corresponds to selecting a subset of arms, and the number of feasible experiments grows on the order of N^K . Moreover, even for a fixed subset, the information induced by the MNL model is fundamentally different from that in linear or generalized linear models. In particular, the Fisher information is matrix-valued, depends nonlinearly on the underlying parameter, and is determined by the entire subset through MNL choice probabilities, rather than decomposing additively across individual arms. As a result, classical optimal design theory (Kiefer & Wolfowitz, 1960; Pukelsheim, 2006)—which relies on additive, per-arm information contributions—does not directly apply.

In this paper, we develop an optimal design framework for MNL models with combinatorial experiments that addresses the structural and computational challenges described above. Our approach builds on *locally* optimal design principles to handle the inherent nonlinearity of the MNL model, and leads to a Frank–Wolfe algorithm (Frank et al., 1956; Dunn & Harshbarger, 1978) whose main computational challenge is a linear maximization oracle (LMO) over a *combinatorial* collection of feasible subsets. To address this bottleneck, we propose two complementary approaches: (i) an *exact* 0–1 mixed-integer linear program (MILP) reformulation of the LMO, allowing *early stopping* with guaranteed accuracy up to a user-specified tolerance, and (ii) a *polynomial-time* relaxation based on a Schur-complement lifting of the MNL Fisher information with a bounded approximation error.

As an application of our efficient optimal design framework, we study *best assortment identification* in MNL bandits with linear utilities. In this fixed-confidence setting, an

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agent sequentially offers assortments (i.e., subsets of arms) and observes a single MNL choice outcome, with the goal of identifying the unique revenue-maximizing assortment S^* with high probability while minimizing the total number of interactions. Leveraging a G -optimal design computed at a nominal parameter, we propose an algorithm, BSI-MNL, that combines an initial exploration phase (which ensures the nominal parameter is uniformly close to the true parameter), design-based sampling, and a principled stopping rule. We prove an instance-dependent sample complexity bound of $\tilde{O}(d \log(N/\delta)(\Delta_{\min}^{-2} + (\kappa \Delta_{\min})^{-1}))$, which in the small-gap regime simplifies to the leading-order guarantee $\tilde{O}(\frac{d \log N}{\Delta_{\min}^2})$. To the best of our knowledge, this yields the first sample complexity guarantee for best assortment identification in MNL bandits with linear utilities, while remaining computationally efficient via either MILP-based early stopping or the polynomial-time lifted relaxation.

Our main contributions are summarized as follows:

- **Exact MILP reformulation:** To address the primary computational bottleneck of the Frank–Wolfe (FW) algorithm—the linear maximization oracle (LMO) defined in (3)—we show that the optimization problem admits an *exact* reformulation as a 0–1 mixed-integer linear program (Theorem 3.3). This reformulation allows the LMO to be solved using off-the-shelf MILP solvers, which either return an exact solution or terminate early with a user-specified, solver-certified optimality gap. Moreover, we show that the FW algorithm retains its ϵ -optimality guarantee even with certified inexact LMO solutions, at the cost of a small additional number of FW iterations (Proposition 3.4).
- **Polynomial-time lifted surrogate:** As an alternative approach, we propose a novel lifted surrogate based on a Schur-complement construction. This relaxation yields a ratio-of-sums objective that can be optimized *exactly* in polynomial time. The resulting design trades statistical efficiency for computational scalability, while incurring only a bounded and explicitly characterized approximation error (Theorem 3.5).
- **Best assortment identification in MNL bandits:** Leveraging the optimal design, we propose BSI-MNL for best assortment identification in MNL bandits, and establish an instance-dependent sample complexity of $\tilde{O}(\frac{d \log N}{\Delta_{\min}^2})$, where d is the feature dimension, N is the number of arms, and $\Delta_{\min}$ is the minimum revenue gap (Theorem 4.4). To the best of our knowledge, this is the first sample complexity guarantee for best assortment identification in MNL bandits with linear utilities and non-uniform revenue parameters. Moreover, the algorithm remains computationally efficient, either via MILP-based early stopping with certified gaps or via a

fully polynomial-time lifted relaxation.

2. Related Work

MNL model in Bandits. The MNL feedback model has been extensively studied in the bandit literature (Agrawal et al., 2019; 2017; Ou et al., 2018; Chen et al., 2020; Oh & Iyengar, 2019; 2021; Perivier & Goyal, 2022; Agrawal et al., 2023; Lee & Oh, 2024). Compared to linear, generalized linear, and logistic bandits (Abbasi-Yadkori et al., 2011; Faury et al., 2020; Abeille et al., 2021; Faury et al., 2022; Lee et al., 2024), MNL bandits present additional challenges due to their combinatorial action space and non-uniform revenue parameters. Most prior work primarily focuses on regret minimization (Chen et al., 2020; Oh & Iyengar, 2019; 2021; Perivier & Goyal, 2022; Lee & Oh, 2024; 2025). Although a limited number of studies address best-arm or best-assortment identification (Rusmevichientong et al., 2010; Sauré & Zeevi, 2013; Yang, 2021; Karpov & Zhang, 2022), they are largely restricted to context-free MNL settings. Consequently, best assortment identification in MNL bandits with linear utilities remains underexplored.

Pure Exploration in Linear Bandits. The pure exploration problem has been extensively investigated in linear bandits (Soare et al., 2014; Degenne et al., 2020; Li et al., 2022; Jedra & Proutiere, 2020; Fiez et al., 2019; Komiya et al., 2022; Yang & Tan, 2022). In this line of work, G -optimal design plays a central role in achieving near-optimal sample complexity. These approaches aim to identify the best arm or to certify near-optimality with high confidence, yielding strong theoretical and practical guarantees. However, existing design-based methods rely on enumerating individual arms and therefore do not extend to MNL bandits, where the action space is inherently combinatorial. Consequently, prior work provides no sample-complexity guarantees for pure exploration in MNL bandits.

3. Optimal Design for MNL Model

3.1. Notations.

For a positive integer n , we define $[n]$ as the set $\{1, 2, \dots, n\}$. The ℓ_2 -norm of a vector x is denoted by $\|x\|_2$. For a positive semi-definite matrix A and a vector x , we use $\|x\|_A$ to represent $\sqrt{x^\top A x}$. For any two symmetric matrices A and B of the same dimensions, $A \geq B$ indicates that $A - B$ is a positive semi-definite matrix. Finally, we define $\mathcal{S}$ as the set of candidate assortments with a size constraint of at most K , i.e., $\mathcal{S} = \{S \subseteq [N] : |S| \leq K\}$.

3.2. MNL Model and Locally Optimal Design

MNL model and loss function. We consider the problem of optimal experimental design under the multinomial logit

(MNL) model (McFadden, 1977). Let $\mathcal{A} = \{\mathbf{a}_i\}_{i=1}^N$ denote a finite set of N arms, where each arm $i \in [N]$ is represented by a d -dimensional feature vector $\mathbf{a}_i \in \mathbb{R}^d$. Let $\mathcal{S}$ denote the collection of feasible subsets of arms with maximum size K . At each step t , the learner selects a subset $S_t \in \mathcal{S}$ with $2 \leq |S_t| \leq K$.¹ Conditional on S_t , the probability of choosing arm $i \in S_t$ follows the MNL model:

$$p(i|S_t, \boldsymbol{\theta}^*) := \frac{\exp(\mathbf{a}_i^\top \boldsymbol{\theta}^*)}{\sum_{j \in S_t} \exp(\mathbf{a}_j^\top \boldsymbol{\theta}^*)}, \quad (1)$$

where $\boldsymbol{\theta}^* \in \mathbb{R}^d$ is an *unknown* underlying parameter. Note that when $K = 2$, the MNL model reduces to the Bradley–Terry–Luce (BTL) model (Bradley & Terry, 1952).

At step t , let $\mathbf{y}_t := (y_{ti})_{i \in S_t} \in \{0, 1\}^{|S_t|}$ denote the one-hot choice feedback vector, where $y_{ti} = 1$ indicates that arm $i \in S_t$ is selected. Then $\Pr(y_{ti} = 1|S_t) = p(i|S_t, \boldsymbol{\theta}^*)$, where $i \in S_t$, and $\sum_{i \in S_t} y_{ti} = 1$.

Now, we define the MNL loss function at step t as:

$$\ell_t(\boldsymbol{\theta}) := - \sum_{i \in S_t} y_{ti} \log p(i|S_t, \boldsymbol{\theta}).$$

Then, the Fisher information matrix (Fisher, 1925) is defined as the expected Hessian of the negative log-likelihood:

$$\begin{aligned} \mathbf{I}_\theta(S_t) &:= \mathbb{E}_{\mathbf{y}_t} [\nabla^2 \ell_t(\boldsymbol{\theta}) | S_t] = \nabla^2 \ell_t(\boldsymbol{\theta}) \\ &= \sum_{i \in S_t} p(i|S_t, \boldsymbol{\theta}) (\mathbf{a}_i - \bar{\mathbf{a}}_\theta(S_t)) (\mathbf{a}_i - \bar{\mathbf{a}}_\theta(S_t))^\top, \end{aligned}$$

where $\bar{\mathbf{a}}_\theta(S_t) := \sum_{j \in S_t} p(j|S_t, \boldsymbol{\theta}) \mathbf{a}_j$ denotes the mean feature vector under the MNL choice probabilities. The second equality holds because, under the MNL model, the Hessian of the negative log-likelihood, $\nabla^2 \ell_t(\boldsymbol{\theta})$, does not depend on the *realized* choice feedback.

Throughout the paper, we assume that the pairwise feature differences span the d -dimensional space:

Assumption 3.1 (Spanning condition). We assume that $\text{span}\{\mathbf{a}_i - \mathbf{a}_j : i, j \in [N], i \neq j\} = \mathbb{R}^d$.

This assumption holds *without loss of generality*: if the span were a strict subspace of $\mathbb{R}^d$, we could restrict $\boldsymbol{\theta}$ to the identifiable subspace (and redefine d accordingly), so that Assumption 3.1 holds throughout, without changing the MNL probabilities (see Appendix A for more details).

Locally G-optimal design. We focus on solving the *G-optimal design* problem (Kiefer & Wolfowitz, 1960; Pukelsheim, 2006) under the MNL model, in which the set of feasible assortments is combinatorial and has cardinality on the order of $\mathcal{O}(N^K)$. Let $\pi : \mathcal{S} \rightarrow [0, 1]$ denote a

probability distribution over feasible assortments. Given any fixed parameter $\boldsymbol{\theta}_0 \in \mathbb{R}^d$ —hence the term *locally* optimal design—our goal is to find $\pi_{\boldsymbol{\theta}_0}^* \in \arg\min_{\pi} g_{\boldsymbol{\theta}_0}(\pi)$, where

$$g_{\boldsymbol{\theta}_0}(\pi) = \max_{S \in \mathcal{S}} \text{tr}(\mathbf{M}_{\boldsymbol{\theta}_0}(\pi)^{-1} \mathbf{I}_{\boldsymbol{\theta}_0}(S)). \quad (2)$$

Here, $\mathbf{M}_{\boldsymbol{\theta}_0}(\pi) := \mathbb{E}_{S \sim \pi} [\mathbf{I}_{\boldsymbol{\theta}_0}(S)]$ denotes the expected Fisher information matrix induced by the design π . Note that, since the objective in (2) does not depend on the realized feedback, algorithms based on the optimal design $\pi_{\boldsymbol{\theta}_0}^*$ are non-adaptive.

It is well known that solutions to the optimal design problem are characterized by the Kiefer–Wolfowitz (KW) equivalence theorem (Kiefer & Wolfowitz, 1960; Lattimore & Szepesvári, 2020). However, whereas the classical KW theorem is formulated in terms of feature vectors, we consider a matrix-valued information form $\mathbf{I}_{\boldsymbol{\theta}_0}(S) \in \mathbb{R}^{d \times d}$. Therefore, we establish the KW equivalence theorem for the MNL model, building on the matrix-valued KW theorem of Mukherjee et al. 2024, Theorem 1.

Proposition 3.2 (KW equivalence theorem for MNL). *Under Assumption 3.1, the following statements are equivalent:*

- (a) $\pi_{\boldsymbol{\theta}_0}^*$ is a minimizer of $g_{\boldsymbol{\theta}_0}(\pi)$, defined in (2).
- (b) $\pi_{\boldsymbol{\theta}_0}^*$ is a maximizer of $f_{\boldsymbol{\theta}_0}(\pi) = \log \det(\mathbf{M}_{\boldsymbol{\theta}_0}(\pi))$.
- (c) $g_{\boldsymbol{\theta}_0}(\pi_{\boldsymbol{\theta}_0}^*) = d$.

Furthermore, there exists a minimizer $\pi_{\boldsymbol{\theta}_0}^*$ of $g_{\boldsymbol{\theta}_0}(\pi)$ such that $|\text{supp}(\pi_{\boldsymbol{\theta}_0}^*)| \leq d(d+1)/2$.

Proposition 3.2 shows the equivalence between G-optimality and D-optimality under the MNL model, and further guarantees the existence of an optimal design with finite support. The proof of Proposition 3.2 is provided in Appendix B.3.

3.3. Frank-Wolfe Algorithm for MNL model

To solve (2), we employ a Frank–Wolfe algorithm (Frank et al., 1956; Nocedal, 2006; Jaggi, 2013), tailored to the MNL model (see Algorithm 1). Rather than directly minimizing the G-optimality objective $g_{\boldsymbol{\theta}_0}(\pi)$, we exploit its equivalent D-optimal formulation at the nominal parameter $\boldsymbol{\theta}_0$, and instead maximize $f_{\boldsymbol{\theta}_0}(\pi) := \log \det(\mathbf{M}_{\boldsymbol{\theta}_0}(\pi))$ over $\pi \in \Delta(\mathcal{S})$ (see Proposition 3.2).

In Algorithm 1, we fix a nominal parameter $\boldsymbol{\theta}_0$ and initialize π_0 such that $\mathbf{M}_{\boldsymbol{\theta}_0}(\pi_0) > 0$. At iteration m , let π_m denote the current design and define $\mathbf{M}_m := \mathbf{M}_{\boldsymbol{\theta}_0}(\pi_m)$. Using the matrix derivative $\nabla_{\mathbf{M}} \log \det(\mathbf{M}) = \mathbf{M}^{-1}$ together with the chain rule, the partial derivative of $f_{\boldsymbol{\theta}_0}(\pi_m)$ with respect to the coordinate $\pi(S)$ is given by

$$[\nabla_{\pi} f_{\boldsymbol{\theta}_0}(\pi_m)]_S = \langle \mathbf{M}_m^{-1}, \mathbf{I}_{\boldsymbol{\theta}_0}(S) \rangle = \text{tr}(\mathbf{M}_m^{-1} \mathbf{I}_{\boldsymbol{\theta}_0}(S)).$$

¹For a singleton subset, the choice probability is identically 1; hence, we exclude this trivial case in this section.

Algorithm 1 Frank–Wolfe Algorithm for MNL at θ_0

- 1: **Input:** nominal parameter θ_0 , initial distribution $\pi_0 \in \Delta(\mathcal{S})$, FW precision ε .
- 2: **Initialize:** $\mathbf{M}_0 = \mathbf{M}_{\theta_0}(\pi_0)$, $m = 0$.
- 3: **while** $g_{\theta_0}(\pi_m) > (1 + \varepsilon)d$ **do**
- 4: $S_m \leftarrow \operatorname{argmax}_{S \in \mathcal{S}} \operatorname{tr}(\mathbf{M}_m^{-1} \mathbf{I}_{\theta_0}(S))$.
- 5: $\gamma_m \leftarrow \operatorname{argmax}_{\gamma \in [0,1]} f_{\theta_0}((1 - \gamma)\pi_m + \gamma \mathbb{1}_{S_m})$.
- 6: $\pi_{m+1} = (1 - \gamma_m)\pi_m + \gamma_m \mathbb{1}_{S_m}$.
- 7: Update $\mathbf{M}_{m+1}$ and $m \leftarrow m + 1$.
- 8: **end while**
- 9: Output the estimated policy $\hat{\pi}_{\theta_0} = \pi_m$.

The *linear maximization oracle* (LMO) therefore, selects an extreme point $\mathbb{1}_{S_m} \in \Delta(\mathcal{S})$, where $\mathbb{1}_S$ denotes the Dirac distribution at S , corresponding to

$$S_m \in \operatorname{argmax}_{S \in \mathcal{S}} \operatorname{tr}(\mathbf{M}_m^{-1} \mathbf{I}_{\theta_0}(S)). \quad (3)$$

We then update the design by taking a convex combination of the current iterate π_m and the extreme point $\mathbb{1}_{S_m}$, i.e., $\pi_{m+1} = (1 - \gamma_m)\pi_m + \gamma_m \mathbb{1}_{S_m}$, where the step-size $\gamma_m \in [0, 1]$ is selected via a line search that maximizes $f_{\theta_0}((1 - \gamma)\pi_m + \gamma \mathbb{1}_{S_m})$. It is known that this procedure converges to a maximizer of $f_{\theta_0}(\pi)$ (Zhao & Freund, 2023).² We refer to the returned policy $\hat{\pi}_{\theta_0}$ as a $(1 + \varepsilon)$ -approximation of the optimal policy $\pi_{\theta_0}^*$.

Computational challenge in LMO. However, a naive implementation of Algorithm 1 is computationally infeasible, as the LMO in (3) entails exhaustive enumeration over the set of all assortments, whose cardinality is $|\mathcal{S}| = \mathcal{O}(N^K)$.

In the following two subsections, we address this computational bottleneck in two complementary ways. **First**, we reformulate (3) as a 0–1 *mixed-integer linear program* (MILP), which enables the LMO to be solved using standard MILP solvers with early stopping and solver-certified optimality gaps (Subsection 3.4). **Second**, we relax the original problem in (3) via a *Schur-complement lifting*, resulting in a polynomial-time surrogate LMO at the cost of approximation error; importantly, the resulting worst-case error admits a provable upper bound (Subsection 3.5).

3.4. LMO via 0–1 MILP with Certified Optimality Gaps

In this subsection, we show that the linear minimization oracle (LMO) in (3) can be reformulated as a 0–1 *mixed-integer linear program* (MILP).

Theorem 3.3 (MILP reformulation of the LMO). *The optimization problem in (3) admits an exact reformulation as a compact 0–1 mixed-integer linear program (MILP) whose*

²Since inexact line search is standard in Frank–Wolfe methods and the resulting error is typically negligible in practice, we omitted this approximation error from the presentation for simplicity.

number of variables and linear constraints is polynomial in N . As a consequence, the LMO can be solved using standard off-the-shelf MILP solvers either to global optimality or, via early termination, with a solver-certified optimality gap at most ϵ_{LMO} .

The proof of Theorem 3.3 is deferred to Appendix B.1.

Early stopping with ϵ_{LMO} -optimality. Although the resulting MILP is NP-hard in general, modern MILP solvers based on branch-and-bound maintain both (i) the best feasible solution found so far and (ii) a provable bound on how good any solution can possibly be. From these two quantities, the solver reports a *certified optimality gap* at runtime, which upper-bounds the remaining suboptimality of the current incumbent solution (the best feasible solution). Therefore, the LMO computation can be safely terminated once this solver-certified gap drops below a user-specified tolerance ϵ_{LMO} . This yields an *instance-dependent* and *verifiable early-stopping* rule: the returned LMO solution comes with an explicit certificate of near-optimality, providing a principled trade-off between computational cost and user-specified optimization accuracy.

Frank–Wolfe stopping under ϵ_{LMO} -approximate LMO. When the MILP-based LMO is solved approximately, the solver returns a candidate set $\hat{S}_m$ together with a certified optimality gap. To account for this inexactness, we tighten the Frank–Wolfe stopping rule. Specifically, if the solver is terminated once the certified gap is at most ϵ_{LMO} , we define

$$\tilde{\epsilon} := \epsilon - \epsilon_{\text{LMO}}/d,$$

and require $\tilde{\epsilon} > 0$. The FW algorithm is then run with tolerance $\tilde{\epsilon}$. See Algorithm B.1 for a complete description.

Proposition 3.4 (FW stopping guarantee under an inexact LMO). *Let $\tilde{\epsilon} \in (0, 1]$. Then the Frank–Wolfe algorithm with tolerance $\tilde{\epsilon}$ terminates after at most $\tau = \tilde{\mathcal{O}}(d/\tilde{\epsilon})$ iterations, and returns a design $\hat{\pi}_{\theta_0}$ that is ϵ -accurate.*

Proposition 3.4 shows that, even with an ϵ_{LMO} -approximate LMO, the FW algorithm terminates in at most $\tilde{\mathcal{O}}(d/\tilde{\epsilon})$ iterations and returns an ϵ -accurate design. The proof of Proposition 3.4 is deferred to Appendix B.4.

Comparison with Thekumparampil et al. 2024. Recently, Thekumparampil et al. (2024) proposed a randomized FW algorithm (DopeWolfe) that approximates the LMO by sampling a subset of candidate assortments $\mathcal{R}_m \subseteq \mathcal{S}$ at each iteration m . However, to guarantee a small Frank–Wolfe approximation error, the required sample size still scales as $|\mathcal{R}_m| = \mathcal{O}(N^K)$ (see Corollary 4 in Thekumparampil et al., 2024), and thus the combinatorial nature of the LMO is not avoided. Moreover, unlike our MILP-based approach, DopeWolfe requires a *pre-specified* iteration budget and does not admit a solver-certified early-stopping rule.

3.5. Polynomial-Time Surrogate LMO via Schur-Complement Lifting

In this subsection, we develop a lifting-based reformulation of the MNL Fisher information that removes the set-dependent centering term $\bar{\mathbf{a}}_S(\boldsymbol{\theta})$ and enables an efficient surrogate linear maximization oracle (LMO). Throughout, we fix a nominal parameter $\boldsymbol{\theta}_0 \in \mathbb{R}^d$ and write $p_{\boldsymbol{\theta}_0}(i|S) := p(i|S, \boldsymbol{\theta}_0)$ for brevity.

Schur-complement lifting. For each arm $i \in [N]$, define the lifted feature vector $\tilde{\mathbf{a}}_i := (\mathbf{a}_i^\top, 1)^\top \in \mathbb{R}^{d+1}$ and, for each assortment $S \in \mathcal{S}$, define the *lifted* (uncentered) second-moment matrix

$$\tilde{\mathbf{I}}_{\boldsymbol{\theta}_0}(S) := \sum_{i \in S} p_{\boldsymbol{\theta}_0}(i|S) \tilde{\mathbf{a}}_i \tilde{\mathbf{a}}_i^\top \in \mathbb{R}^{(d+1) \times (d+1)}. \quad (4)$$

Let $\bar{\mathbf{A}}_{\boldsymbol{\theta}_0}(S) := \sum_{i \in S} p_{\boldsymbol{\theta}_0}(i|S) \mathbf{a}_i \mathbf{a}_i^\top$. Then $\tilde{\mathbf{I}}_{\boldsymbol{\theta}_0}(S)$ defined in (4) admits the block decomposition

$$\tilde{\mathbf{I}}_{\boldsymbol{\theta}_0}(S) = \begin{bmatrix} \bar{\mathbf{A}}_{\boldsymbol{\theta}_0}(S) & \bar{\mathbf{a}}_{\boldsymbol{\theta}_0}(S) \\ \bar{\mathbf{a}}_{\boldsymbol{\theta}_0}(S)^\top & 1 \end{bmatrix}, \quad (5)$$

and the (centered) MNL Fisher information satisfies $\mathbf{I}_{\boldsymbol{\theta}_0}(S) = \bar{\mathbf{A}}_{\boldsymbol{\theta}_0}(S) - \bar{\mathbf{a}}_{\boldsymbol{\theta}_0}(S) \bar{\mathbf{a}}_{\boldsymbol{\theta}_0}(S)^\top$, which is precisely the Schur complement of the bottom-right block of (5).³ In particular, the dependence of $\mathbf{I}_{\boldsymbol{\theta}_0}(S)$ on the centering term $\bar{\mathbf{a}}_{\boldsymbol{\theta}_0}(S)$ is captured linearly in the lifted matrix $\tilde{\mathbf{I}}_{\boldsymbol{\theta}_0}(S)$.

Given a design $\pi \in \Delta(\mathcal{S})$, we define the lifted design matrix

$$\tilde{\mathbf{M}}_{\boldsymbol{\theta}_0}(\pi) := \mathbb{E}_{S \sim \pi} [\tilde{\mathbf{I}}_{\boldsymbol{\theta}_0}(S)] = \begin{bmatrix} \bar{\mathbf{B}}_{\boldsymbol{\theta}_0}(\pi) & \bar{\mathbf{b}}_{\boldsymbol{\theta}_0}(\pi) \\ \bar{\mathbf{b}}_{\boldsymbol{\theta}_0}(\pi)^\top & 1 \end{bmatrix},$$

where $\bar{\mathbf{B}}_{\boldsymbol{\theta}_0}(\pi) := \mathbb{E}_\pi [\bar{\mathbf{A}}_{\boldsymbol{\theta}_0}(S)]$ and $\bar{\mathbf{b}}_{\boldsymbol{\theta}_0}(\pi) := \mathbb{E}_\pi [\bar{\mathbf{a}}_{\boldsymbol{\theta}_0}(S)]$. Moreover, we define its Schur-complement matrix

$$\widehat{\mathbf{M}}_{\boldsymbol{\theta}_0}(\pi) := \bar{\mathbf{B}}_{\boldsymbol{\theta}_0}(\pi) - \bar{\mathbf{b}}_{\boldsymbol{\theta}_0}(\pi) \bar{\mathbf{b}}_{\boldsymbol{\theta}_0}(\pi)^\top \in \mathbb{R}^{d \times d},$$

and the associated *design-mismatch* matrix

$$\Delta_{\boldsymbol{\theta}_0}(\pi) := \widehat{\mathbf{M}}_{\boldsymbol{\theta}_0}(\pi) - \mathbf{M}_{\boldsymbol{\theta}_0}(\pi).$$

We call $\Delta_{\boldsymbol{\theta}_0}(\pi)$ a *mismatch* matrix because $\widehat{\mathbf{M}}_{\boldsymbol{\theta}_0}(\pi)$ is the Schur complement of the lifted design matrix $\tilde{\mathbf{M}}_{\boldsymbol{\theta}_0}(\pi)$, whereas $\mathbf{M}_{\boldsymbol{\theta}_0}(\pi)$ is the *true* (centered) Fisher design matrix in the original space. Thus, $\Delta_{\boldsymbol{\theta}_0}(\pi)$ quantifies the discrepancy between the Schur-complement surrogate induced by lifting and the original design matrix.

Lifted G-optimal design. The lifting in (4) naturally induces a *lifted* G-optimal design criterion. For any design $\pi \in \Delta(\mathcal{S})$, define

$$\tilde{g}_{\boldsymbol{\theta}_0}(\pi) := \max_{S \in \mathcal{S}} \text{tr}(\tilde{\mathbf{M}}_{\boldsymbol{\theta}_0}(\pi)^{-1} \tilde{\mathbf{I}}_{\boldsymbol{\theta}_0}(S)).$$

³For a block matrix $\begin{pmatrix} A & B \\ B^\top & C \end{pmatrix}$ with C invertible, the *Schur complement* of the block C is defined as $A - BC^{-1}B^\top$.

We then define the *lifted optimal design* as any minimizer $\tilde{\pi}_{\boldsymbol{\theta}_0}^* \in \arg\min_{\pi \in \Delta(\mathcal{S})} \tilde{g}_{\boldsymbol{\theta}_0}(\pi)$. We solve this lifted design problem by running Frank–Wolfe with $\tilde{g}_{\boldsymbol{\theta}_0}$ as the objective. See Algorithm B.2 for a full description.

Polynomial-time lifted LMO. At iteration m of Algorithm B.2, given the current design π_m and $\tilde{\mathbf{M}}_m := \tilde{\mathbf{M}}_{\boldsymbol{\theta}_0}(\pi_m)$, we compute the next FW atom by solving

$$S_m \in \arg\max_{S \in \mathcal{S}} \text{tr}(\tilde{\mathbf{M}}_m^{-1} \tilde{\mathbf{I}}_{\boldsymbol{\theta}_0}(S)). \quad (6)$$

This lifted oracle admits a ratio-of-sums representation and can be solved in polynomial time. Indeed, letting $s_i := \tilde{\mathbf{a}}_i^\top \tilde{\mathbf{M}}_m^{-1} \tilde{\mathbf{a}}_i$ and $w_i := \exp(\mathbf{a}_i^\top \boldsymbol{\theta}_0)$, for every $S \in \mathcal{S}$ we have

$$\text{tr}(\tilde{\mathbf{M}}_m^{-1} \tilde{\mathbf{I}}_{\boldsymbol{\theta}_0}(S)) = \sum_{i \in S} p_{\boldsymbol{\theta}_0}(i|S) s_i = \frac{\sum_{i \in S} w_i s_i}{\sum_{j \in S} w_j}.$$

Consequently, (6) is a classical MNL ratio-of-sums assortment optimization problem and can be reduced to a linear program (LP) and solved in time $\mathcal{O}(\text{poly}(N))$ (e.g., Rusmevichientong et al., 2010; Davis et al., 2014).

Let $\pi_{\boldsymbol{\theta}_0}^* \in \arg\min_{\pi \in \Delta(\mathcal{S})} g_{\boldsymbol{\theta}_0}(\pi)$ be an original G-optimal design. Under the Kiefer–Wolfowitz equivalence (Proposition 3.2), $g_{\boldsymbol{\theta}_0}(\pi_{\boldsymbol{\theta}_0}^*) = d$. Similarly, under the lifted criterion, one typically has $g_{\boldsymbol{\theta}_0}(\tilde{\pi}_{\boldsymbol{\theta}_0}^*) = d + 1$ (reflecting the lifted dimension).

Theorem 3.5 (Lifted FW guarantee). *Fix $\boldsymbol{\theta}_0$ and run Frank–Wolfe (Algorithm 1) to minimize $\tilde{g}_{\boldsymbol{\theta}_0}(\pi)$ using the polynomial-time lifted LMO (6). Let $\hat{\pi}$ be the design returned at the stopping time $\tilde{g}_{\boldsymbol{\theta}_0}(\hat{\pi}) \leq (1 + \epsilon)(d + 1)$, and define $\epsilon_{\text{lift}} := \inf \{\epsilon \geq 0 : \Delta_{\boldsymbol{\theta}_0}(\hat{\pi}) \leq \epsilon \mathbf{M}_{\boldsymbol{\theta}_0}(\hat{\pi})\}$. Then*

$$g_{\boldsymbol{\theta}_0}(\hat{\pi}) \leq 2(1 + \epsilon_{\text{lift}})(1 + \epsilon)d.$$

Computational–statistical trade-off. Using the lifted LMO yields a polynomial-time Frank–Wolfe update. The trade-off is a potential mismatch between the lifted surrogate and the true Fisher information, quantified by ϵ_{lift} . Accordingly, the G-optimality guarantee degrades by at most a factor $(1 + \epsilon_{\text{lift}})$. A crude upper bound is $\mathcal{O}(1/\lambda_{\min}(\mathbf{M}_{\boldsymbol{\theta}_0}(\tilde{\pi}_{\boldsymbol{\theta}_0}^*)))$ (Proposition B.4). In the standard MNL bandit setting with an outside option (see Equation (7)), we also obtain the bound $\epsilon_{\text{lift}} = \mathcal{O}(Ke^B \wedge 1/\lambda_{\min}(\mathbf{M}_{\boldsymbol{\theta}_0}(\tilde{\pi}_{\boldsymbol{\theta}_0}^*)))$, when $\|\boldsymbol{\theta}_0\| \leq B$ (Proposition B.5). The proof of Theorem 3.5 is provided in Appendix B.5.

4. Best Assortment Identification in MNL Bandits with Linear Utilities

In this section, we consider *best assortment identification* (BSI) in MNL bandits, where the abbreviation BSI is used throughout to avoid confusion with best arm identification

(BAI). Following the previous literature (Agrawal et al., 2019; 2017; Ou et al., 2018; Chen et al., 2020; Oh & Iyengar, 2019; 2021; Perivier & Goyal, 2022; Agrawal et al., 2023; Lee & Oh, 2024; 2025), we adopt the MNL choice model with an *outside option* (corresponding to not choosing any item in the offered assortment S), where the choice probabilities are given by

$$p(i|S, \theta^*) := \frac{\exp(\mathbf{a}_i^\top \theta^*)}{1 + \sum_{j \in S} \exp(\mathbf{a}_j^\top \theta^*)}, \quad \forall i \in S, \quad (7)$$

and $p(0|S, \theta^*) := 1 - \sum_{i \in S} p(i|S, \theta^*)$. Here 0 denotes the outside option.

We define the expected revenue of an assortment S as:

$$R(S, \theta) = \sum_{i \in S} p(i|S, \theta) r_i = \frac{\sum_{i \in S} \exp(\mathbf{a}_i^\top \theta) r_i}{1 + \sum_{j \in S} \exp(\mathbf{a}_j^\top \theta)},$$

where $r_i \in [0, 1]$ is a given revenue parameter. Let $S^* \in \arg\max_{S \in \mathcal{S}} R(S, \theta^*)$ denote the optimal assortment and assume the maximizer is unique. Given a failure probability $\delta > 0$, our goal is to interact with the environment until a (possibly random) stopping time τ and return an assortment $\hat{S}$ such that

$$\mathbb{P}(\hat{S} = S^* \wedge \tau < \infty) \geq 1 - \delta.$$

This setting is known as the fixed-confidence setting (Garivier & Kaufmann, 2016), where the objective is to identify the optimal assortment with as few samples as possible.

We work under the standard boundedness assumption.

Assumption 4.1. For all $i \in [N]$, $\|\mathbf{a}_i\|_2 \leq 1$, and $\|\theta^*\|_2 \leq B$.

Moreover, following the previous MNL bandit literature (Oh & Iyengar, 2021; Perivier & Goyal, 2022; Lee & Oh, 2024), we introduce the problem-dependent constant:

$$\kappa := \min_{S \in \mathcal{S}} \min_{i \in S} p(i|S, \theta^*) p(0|S, \theta^*), \quad (8)$$

which measures the degree of deviation from the linear model. Since $1/\kappa = \Theta(K^2 e^{3B})$ grows exponentially in B , it is crucial to derive bounds that do not depend on $1/\kappa$.

Confidence bound. Let $\mathcal{D}_t = \{(i_s, S_s)\}_{s=1}^t$ denote the data collected up to t . We define the cumulative loss as:

$$\mathcal{L}_{\mathcal{D}_t}(\theta) := \sum_{s=1}^t \ell_s(\theta) + \frac{\lambda}{2} \|\theta\|_2^2, \quad \text{where } \lambda > 0.$$

We also define Hessian matrix at θ as $\mathbf{H}_t(\theta) = \mathbf{H}_{\mathcal{D}_t}(\theta) := \sum_{s=1}^t \nabla^2 \ell_s(\theta) + \lambda \mathbf{I}_d$. The following lemma provides a confidence bound of order $\sqrt{\log N}$, independent of d .

Lemma 4.2 (Theorem 1 of Han et al. 2026). *Given $\mathcal{D} := \{(i_t, S_t)\}_{t=1}^T$, let $\hat{\theta}_{\mathcal{D}} \in \arg\min_{\theta \in \mathbb{R}^d} \mathcal{L}_{\mathcal{D}}(\theta)$. Assume that,*

conditional on $\{S_t\}_{t \in [T]}$, the observed choices $\{i_t\}_{t \in [T]}$ are mutually independent and that Assumption 4.1 holds. Define $\mathbf{H}_{\mathcal{D}}(\theta) := \sum_{t \in [T]} \nabla^2 \ell_t(\theta) + \lambda \mathbf{I}_d$. Suppose that the event

$$64 \max_{s \leq t, i \in S_s} \|\mathbf{a}_i\|_{\mathbf{H}_{\mathcal{D}}(\theta^*)^{-1}} \leq \frac{1}{\sqrt{d \log(N/\delta)}} \wedge \frac{1}{\sqrt{\lambda} B} \quad (9)$$

holds. Let $\beta(\delta) := (36\sqrt{\log(N/\delta)} + 64\sqrt{\lambda} B)$. Then, with probability at least $1 - \delta$, for all $i \in [N]$ we have

$$|\mathbf{a}_i^\top (\hat{\theta}_{\mathcal{D}} - \theta^*)| \leq \beta(\delta) \|\mathbf{a}_i\|_{\mathbf{H}_{\mathcal{D}}(\theta^*)^{-1}}.$$

4.1. Algorithm: BSI-MNL

We now describe our algorithm for BSI (Algorithm 2).

Step 1: Initial exploration. In Algorithm 2, we begin with a pure exploration phase that continues until the burn-in condition of Lemma 4.2 is met. The purpose of this phase is to ensure that the uncertainty term is uniformly controlled over all arms. Since the true parameter θ^* is unknown, we cannot directly verify the condition in (9). Instead, we use the fact that $\mathbf{H}_t(\theta^*) \geq \kappa \mathbf{V}_t$, where $\mathbf{V}_t := \sum_{s=1}^t \sum_{i \in S_s} \mathbf{a}_i \mathbf{a}_i^\top + \lambda \mathbf{I}_d$. Therefore, it suffices to ensure that $\kappa^{-1/2} \max_{i \in [N]} \|\mathbf{a}_i\|_{\mathbf{V}_t^{-1}}$ is uniformly bounded:

$$\max_{i \in [N]} \|\mathbf{a}_i\|_{\mathbf{V}_t^{-1}} \leq \frac{\kappa^{1/2}}{256} \left(\frac{1}{\sqrt{d \log(N/\delta)}} + \frac{1}{\sqrt{\lambda} B} \right) =: \zeta_w.$$

Step 2: Fixing nominal parameter. After the initial exploration phase, we estimate a nominal parameter θ_0 using an *independent* feedback dataset $\mathcal{D}'_w$. During the initial exploration, each selected assortment S_t is offered twice, yielding two independent choice outcomes i_t and i'_t . These outcomes are stored in two separate datasets, $\mathcal{D}_w := \{(S_s, i_s)\}_{s=1}^{t_w}$ and $\mathcal{D}'_w := \{(S_s, i'_s)\}_{s=1}^{t_w}$, which are conditionally independent given the offered assortments. We then compute the nominal parameter as the regularized MLE, i.e., $\theta_0 \in \arg\min_{\theta \in \mathbb{R}^d} \mathcal{L}_{\mathcal{D}'_w}(\theta)$. By construction, θ_0 depends only on $\mathcal{D}'_w$ and is therefore independent of the main dataset $\mathcal{D}_w$ used in the subsequent phase. This independence is essential for satisfying the conditional independence requirement in Lemma 4.2 when we later sample assortments according to the design computed at θ_0 .

Finally, by running the warm-up phase for a sufficient number of rounds, we ensure that the nominal parameter θ_0 is uniformly close to θ^* , which in turn implies that $p(i|S, \theta_0) \approx p(i|S, \theta^*)$ for all S and i (see Equation C.10).

Step 3: Computing a G -optimal design at θ_0 . Given the nominal parameter θ_0 , we compute a (near) G -optimal design $\hat{\pi}_{\theta_0}$ by running the Frank-Wolfe algorithm (Algorithm 1). The design can be computed either using a *MILP solver* (Subsection 3.4) or via a relaxed but polynomial-time *lifted G -optimal design* formulation (Subsection 3.5).

Algorithm 2 BSI-MNL(Best Assortment Identification in MNL bandits)

- 1: **Input:** failure level δ , warm-up threshold ζ_w , regularization λ , problem-dependent constant κ , precision ε and ϵ_{LMO} .
- 2: **Initialize:** $t = 1$, $\mathcal{D}_w = \mathcal{D}'_w = \emptyset$, $\mathbf{V}_0 = \lambda \mathbf{I}_d$.
- 3: **while** $\max_{i \in [N]} \|\mathbf{a}_i\|_{\mathbf{V}_{t-1}^{-1}} \leq \zeta_w$ **do**
- 4: Select an arbitrary assortment S_t that contains item i such that $\|\mathbf{a}_i\|_{\mathbf{V}_{t-1}^{-1}} > \zeta_w$.
- 5: Offer S_t twice and observe two independent choice feedbacks i_t, i'_t .
- 6: Update $\mathcal{D}_w \leftarrow \mathcal{D}_w \cup \{(S_t, i_t)\}$, $\mathcal{D}'_w \leftarrow \mathcal{D}'_w \cup \{(S_t, i'_t)\}$, $\mathbf{V}_t \leftarrow \mathbf{V}_{t-1} + \sum_{i \in S_t} \mathbf{a}_i \mathbf{a}_i^\top$, and $t \leftarrow t + 1$.
- 7: **end while**
- 8: Compute $\theta_0 \leftarrow \operatorname{argmin}_{\theta \in \mathbb{R}^d} \mathcal{L}_{\mathcal{D}'_w}(\theta)$ and $\hat{\theta}_{t-1} \leftarrow \operatorname{argmin}_{\theta \in \mathbb{R}^d} \mathcal{L}_{\mathcal{D}_w}(\theta)$.
- 9: Construct a locally optimal design $\hat{\pi}_{\theta_0}$ at θ_0 by running Algorithm B.1 (or Algorithm B.2).
- 10: Set $\mathcal{D}_{t-1} \leftarrow \mathcal{D}_w$ and initialize $\mathbf{H}_{t-1}(\theta_0) \leftarrow \mathbf{H}_{\mathcal{D}_w}(\theta_0)$.
- 11: **while** (10) is not true **do**
- 12: $S_t \sim \hat{\pi}_{\theta_0}$ and observe choice feedback i_t .
- 13: Update $\mathcal{D}_t \leftarrow \mathcal{D}_{t-1} \cup \{(S_t, i_t)\}$ and $\mathbf{H}_t(\theta_0) \leftarrow \mathbf{H}_{t-1}(\theta_0) + \nabla^2 \ell_t(\theta_0)$.
- 14: Compute $\hat{\theta}_t \leftarrow \operatorname{argmin}_{\theta \in \mathbb{R}^d} \mathcal{L}_{\mathcal{D}_t}(\theta)$, and update $t \leftarrow t + 1$.
- 15: **end while**
- 16: **Return:** S_t^{best}

Step 4: Main rounds and stopping rule. Given the design $\hat{\pi}_{\theta_0}$, we run the main rounds by repeatedly sampling assortments $S_t \sim \hat{\pi}_{\theta_0}$ and observing the corresponding choice feedback. At each time t , we form the optimistic ($u_{t,i}^+$) and pessimistic ($u_{t,i}^-$) utilities

$$u_{t,i}^\pm = \mathbf{a}_i^\top \hat{\theta}_t \pm \sqrt{2} \beta(\delta) \|\mathbf{a}_i\|_{\mathbf{H}_t(\theta_0)^{-1}},$$

and define the corresponding optimistic/pessimistic expected revenues for any $S \in \mathcal{S}$ as

$$\tilde{R}_t(S) := \frac{\sum_{i \in S} e^{u_{t,i}^+} r_i}{1 + \sum_{j \in S} e^{u_{t,j}^+}}, \quad \check{R}_t(S) := \frac{\sum_{i \in S} e^{u_{t,i}^-} r_i}{1 + \sum_{j \in S} e^{u_{t,j}^-}}.$$

We then select the pessimistic best assortment and the optimistic best alternative as

$$S_t^{\text{best}} \in \operatorname{argmax}_{S \in \mathcal{S}} \check{R}_t(S), \quad S_t^{\text{alt}} \in \operatorname{argmax}_{S \in \mathcal{S}: S \neq S_t^{\text{best}}} \tilde{R}_t(S).$$

Remark 4.3 (Polynomial-time assortment selection). Note that each of these optimization problems can be solved in polynomial time in N (Rusmevichientong et al., 2010; Davis et al., 2014), analogously to (6). The only subtlety is the constraint $S \neq S_t^{\text{best}}$ in the definition of S_t^{alt} . This can be enforced exactly with at most K additional polynomial-time solves: for each item $i \in S_t^{\text{best}}$, solve the same maximization problem under the extra constraint $i \notin [N]$, and then take the best solution among these K candidates.

Finally, we stop at the first time τ such that

$$\tilde{R}_\tau(S_\tau^{\text{best}}) > \tilde{R}_\tau(S_\tau^{\text{alt}}), \quad (10)$$

and output S_τ^{best} . In Lemma C.7, we show that the above stopping rule is sufficient to guarantee $S_\tau^{\text{best}} = S^*$.

4.2. Sample Complexity

We define the (instance-dependent) expected revenue gap

$$\Delta_{\min} := R(S^*, \theta^*) - \max\{R(S, \theta^*) : S \in \mathcal{S}, S \neq S^*\}.$$

We now establish an instance-dependent upper bound on the sample complexity of Algorithm 2.

Theorem 4.4 (Sample complexity, MILP solver). *Let $\delta \in (0, 1]$, $\lambda = 1$, and $T_0 = \tilde{\Omega}(\kappa^{-1} d^2 \log(N/\delta))$. Suppose that Assumption 4.1 holds. Then, with probability at least $1 - \delta$, Algorithm 2, when implemented with the MILP solver described in Subsection 3.4, returns the optimal assortment S^* , and the stopping time τ satisfies*

$$\tau = \tilde{O}\left(d \log(N/\delta) \left(\frac{1}{\Delta_{\min}^2} + \frac{1}{\kappa \Delta_{\min}}\right) + T_0\right).$$

Discussion of Theorem 4.4. To the best of our knowledge, this is the first sample complexity bound for best assortment identification in MNL bandits, particularly in the presence of non-uniform revenue parameters. For sufficiently small $\Delta_{\min}$, the sample complexity is dominated by the $\Delta_{\min}^{-2}$ term, yielding the leading-order bound $\tilde{O}(\frac{d \log N}{\Delta_{\min}^2})$. This matches the existing lower bound $\Omega(\frac{d}{\Delta_{\min}^2})$ (Theorem 3 of Gupta et al. 2021), up to a logarithmic factor in N . In particular, in this small-gap regime, the bound is independent of $1/\kappa$, which can be exponentially large, i.e., $1/\kappa = \mathcal{O}(K^2 e^{3B})$. The proof is provided in Appendix C.

Corollary 4.5 (Sample complexity, Lifted G-optimal). *Under the setting as in Theorem 4.4, suppose we employ the lifted G-optimal design described in Subsection 3.5. Then, with probability at least $1 - \delta$, we have*

$$\tau = \tilde{O}\left((1 + \epsilon_{\text{lift}}) d \log(N/\delta) \left(\frac{1}{\Delta_{\min}^2} + \frac{1}{\kappa \Delta_{\min}}\right) + T_0\right),$$

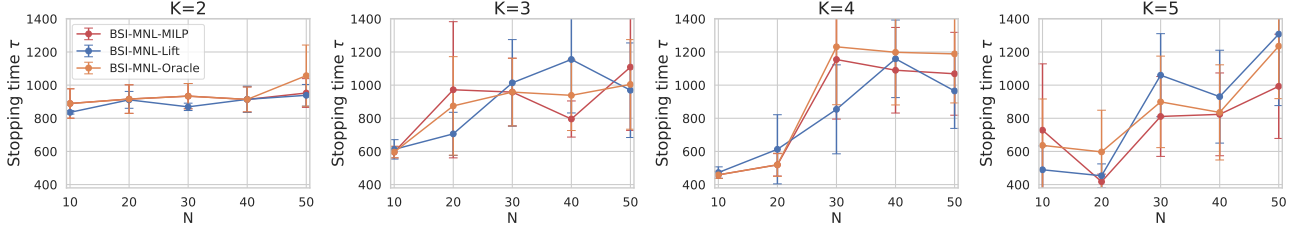


Figure 1. Average stopping time τ for varying N and K over 10 independent random seeds, with one standard deviation.

where ϵ_{lift} denotes the approximation error induced by the lifted G -optimal design (see Theorem 3.5).

Discussion of Corollary 4.5. When we employ the lifted G -optimal design to improve computational efficiency, particularly in large-scale regimes with large N and K , the sample complexity incurs an additional multiplicative factor of $(1 + \epsilon_{\text{lift}})$. By Proposition B.5, this lifting error satisfies $\epsilon_{\text{lift}} = \mathcal{O}(Ke^B)$. For comparison, a naive application of optimal design based on $\mathbf{V}_t$, which ignores local Fisher information and therefore admits only polynomial-time computation (cf. Gupta et al. 2021), yields a sample complexity of $\tilde{\mathcal{O}}(\frac{d \log N}{\kappa^2 \Delta_{\min}^2})$. Since $1/\kappa^2 = \mathcal{O}(K^4 e^{6B})$, this bound can be substantially worse. In contrast, Corollary 4.5 achieves a better worst-case statistical efficiency, by a factor of $K^3 e^{5B}$, while retaining polynomial-time computation.

Remark 4.6 (Relation to item-level gaps). Our gap $\Delta_{\min}$ is an assortment-level global gap between the optimal and second-best assortments, whereas the Δ_i quantities in Yang 2021 and Karpov & Zhang 2022 are item-level separation metrics in MAB settings. In particular, Yang 2021 defines Δ_i using the best revenue among assortments constrained to include or exclude item i , for which one has $\Delta_i \geq \Delta_{\min}$; however, their complexity scales as $\tilde{\mathcal{O}}(\sum_{i=1}^N \Delta_i^{-2})$, while ours scales as $\tilde{\mathcal{O}}(\Delta_{\min}^{-2})$, so the two bounds are not directly comparable: theirs uses a finer item-wise hardness measure, whereas ours uses a single global assortment-level gap, and neither uniformly dominates the other based on the gap term alone. Meanwhile, Karpov & Zhang 2022 defines Δ_i through the advantage score $\eta_i = (r_i - \theta_v)v_i$ and its separation from the top- K decision boundary; hence, their gap is not based on constrained assortment revenues, but on item-wise margins relative to the advantage-score boundary, capturing a different notion of hardness from $\Delta_{\min}$ and making term-by-term comparison again inappropriate.

5. Numerical Experiments

We empirically evaluate the performance of BSI-MNL by measuring the sample complexity up to the stopping time. All results are averaged over 10 independent problem instances, and we report the mean together with the standard deviation. In each instance, the true parameter θ^* is drawn

uniformly from the d -dimensional Euclidean ball $\mathbb{B}^d(B)$ of radius B , and the feature vectors $\{\mathbf{a}_i\}_{i=1}^N$ are sampled independently from the unit ball $\mathbb{B}^d(1)$. The revenue parameters are independently sampled as $r_i \sim \text{Unif}(0, 1)$ for all $i \in [N]$.

To implement the Frank–Wolfe (FW) procedure (Algorithm 1) within BSI-MNL, we consider two variants that differ in how the linear minimization oracle (LMO) is computed. The first variant, denoted BSI-MNL-MILP, solves the LMO by reformulating the original combinatorial optimization problem as a mixed-integer linear program and uses the Gurobi solver (Gurobi, 2024), as described in Subsection 3.4. The second variant, denoted BSI-MNL-Lift, replaces the original LMO with a Schur-complement-based lifted surrogate, which admits polynomial-time computation; see Subsection 3.5 for details. Since this work is, to our knowledge, the first to study best assortment identification with provable guarantees, there is no existing algorithm that serves as a direct baseline. Instead, we include an oracle version of our algorithm that computes the LMO exactly via brute-force enumeration, denoted BSI-MNL-Oracle.

In the FW procedure, we use a stopping accuracy of $\epsilon = 0.1$. For the MILP-based variant, BSI-MNL-MILP, the certified optimality gap of the linear minimization oracle is set to $\epsilon_{\text{LMO}} = 0.1$. For the confidence radius $\beta(\delta)$ and the warm-up threshold ζ_w , we use scaled-down versions of their theoretical expressions by a factor of 10, as the original theoretical values are overly conservative in practice and lead to unnecessarily long warm-up phases and delayed stopping times. Throughout, we set $d = 5$, $B = 1$, and $\delta = 0.05$.

Stopping time. Figure 1 reports the stopping time τ of BSI-MNL as a function of the number of items N and the assortment size K . Both proposed methods, BSI-MNL-MILP and BSI-MNL-Lift, achieve stopping times comparable to the oracle implementation BSI-MNL-Oracle. These results demonstrate that our proposed methods retain statistical efficiency while substantially reducing the computational cost associated with brute-force LMO computations.

One interesting observation is that, for larger K , the sample complexity appears to be more sensitive to N . We conjecture that this behavior is partly driven by the revenue gap $\Delta_{\min} = \min_{S \neq S^*} (R(S^*, \theta^*) - R(S, \theta^*))$, which for larger K may decrease more rapidly with N . Indeed, by the defi-

Table 1. Average LMO runtime (seconds) with standard deviation over 10 independent random seeds. The third column reports the combinatorial complexity $\binom{N}{K}$, highlighting the scalability limits of brute-force enumeration. Bold entries denote the fastest and second-fastest methods for each (N, K) configuration. For the brute-force method, entries marked as “–” indicate settings where exhaustive enumeration was computationally infeasible within the allotted time budget (e.g., 10,000 s).

N	K	$\binom{N}{K}$	BSI-MNL-MILP	BSI-MNL-Lift	BSI-MNL-Oracle
30	3	4,060	2.06 (± 0.34)	0.16 (± 0.05)	0.38 (± 0.13)
	4	27,405	12.76 (± 4.48)	0.15 (± 0.04)	2.63 (± 0.68)
	5	142,506	41.02 (± 16.27)	0.18 (± 0.06)	17.02 (± 6.23)
50	3	19,600	9.10 (± 1.07)	0.14 (± 0.04)	1.57 (± 0.34)
	4	230,300	77.67 (± 18.16)	0.17 (± 0.07)	38.20 (± 7.06)
	5	2,118,760	135.14 (± 47.82)	0.16 (± 0.06)	329.23 (± 56.10)
100	3	161,700	121.84 (± 41.27)	0.16 (± 0.07)	21.06 (± 5.44)
	4	3,921,225	135.38 (± 54.01)	0.17 (± 0.08)	368.94 (± 82.40)
	5	75,287,520	214.13 (± 101.56)	0.17 (± 0.08)	9048.44 (± 1579.01)
200	3	1,313,400	132.49 (± 119.84)	0.15 (± 0.05)	149.40 (± 26.50)
	4	64,684,950	271.88 (± 183.39)	0.17 (± 0.09)	7154.85 (± 1395.94)
	5	2,535,650,040	325.60 (± 203.32)	0.18 (± 0.09)	–

dition of $R(S, \theta)$, adding more items to an assortment tends to increase the expected revenue, which in turn can reduce the revenue gaps, especially when many items are available.

Runtime. Table 1 compares the computational cost of solving the LMO, the main computational bottleneck of our method. Owing to its polynomial-time formulation, BSI-MNL-Lift is consistently and significantly faster than the other approaches, with nearly constant runtime across all tested (N, K) configurations. Moreover, for problem instances with large $\binom{N}{K}$, BSI-MNL-MILP exhibits robust computational performance compared to BSI-MNL-Oracle, whose exhaustive enumeration cost grows rapidly and becomes infeasible for $(N, K) = (200, 5)$.

6. Conclusion

In this paper, we studied optimal experimental design for MNL models and proposed two computationally efficient frameworks to address the combinatorial nature of the action space. Building on these design methods, we developed a best assortment identification algorithm for MNL bandits and established an instance-dependent sample complexity.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Justification of Assumption 3.1 (Without Loss of Generality)

In this appendix, we justify that Assumption 3.1 can be imposed without loss of generality. Recall that under the MNL model, for any assortment $S \subseteq [N]$ and parameter $\theta \in \mathbb{R}^d$, the choice probability of item $i \in S$ is defined as follows:

$$p(i|S, \theta) = \frac{\exp(\mathbf{a}_i^\top \theta)}{\sum_{j \in S} \exp(\mathbf{a}_j^\top \theta)}. \quad (\text{A.1})$$

Identifiable subspace. Define the subspace generated by all pairwise feature differences as

$$\mathcal{U} := \text{span} \{ \mathbf{a}_i - \mathbf{a}_j : i, j \in [N], i \neq j \} \subseteq \mathbb{R}^d.$$

If $\mathcal{U} \neq \mathbb{R}^d$, then there exists a nontrivial orthogonal complement $\mathcal{U}^\perp := \{u \in \mathbb{R}^d : v^\top u = 0, \forall v \in \mathcal{U}\}$. For any $u \in \mathcal{U}^\perp$ and any $i, j \in [N]$, we have

$$(\mathbf{a}_i - \mathbf{a}_j)^\top u = 0 \implies \mathbf{a}_i^\top u = \mathbf{a}_j^\top u.$$

Hence, $\mathbf{a}_i^\top u$ is constant over $i \in [N]$. In particular, fixing any $i_0 \in [N]$ and letting $c(u) := \mathbf{a}_{i_0}^\top u$, we obtain

$$\mathbf{a}_i^\top u = c(u), \quad \forall i \in [N], \forall u \in \mathcal{U}^\perp. \quad (\text{A.2})$$

Decomposition of the parameter. Decompose the parameter as $\theta = \theta_{\mathcal{U}} + \theta_{\mathcal{U}^\perp}$, where $\theta_{\mathcal{U}} \in \mathcal{U}$ and $\theta_{\mathcal{U}^\perp} \in \mathcal{U}^\perp$. By (A.2), there exists a scalar $c(\theta_{\mathcal{U}^\perp}) \in \mathbb{R}$ such that

$$\mathbf{a}_i^\top \theta_{\mathcal{U}^\perp} = c(\theta_{\mathcal{U}^\perp}), \quad \forall i \in [N].$$

Therefore, for any $i \in S$,

$$\mathbf{a}_i^\top \theta = \mathbf{a}_i^\top \theta_{\mathcal{U}} + \mathbf{a}_i^\top \theta_{\mathcal{U}^\perp} = \mathbf{a}_i^\top \theta_{\mathcal{U}} + c(\theta_{\mathcal{U}^\perp}), \quad (\text{A.3})$$

where the additive term $c(\theta_{\mathcal{U}^\perp})$ does not depend on i .

Invariance of MNL probabilities. Plugging (A.3) into (A.1), we obtain, for any S and $i \in S$,

$$\begin{aligned} p(i|S, \theta) &= \frac{\exp(\mathbf{a}_i^\top \theta_{\mathcal{U}}) \exp(c(\theta_{\mathcal{U}^\perp}))}{\sum_{j \in S} \exp(\mathbf{a}_j^\top \theta_{\mathcal{U}}) \exp(c(\theta_{\mathcal{U}^\perp}))} \\ &= \frac{\exp(\mathbf{a}_i^\top \theta_{\mathcal{U}})}{\sum_{j \in S} \exp(\mathbf{a}_j^\top \theta_{\mathcal{U}})} = p(i|S, \theta_{\mathcal{U}}). \end{aligned} \quad (\text{A.4})$$

Thus, the MNL choice probabilities depend only on the projection $\theta_{\mathcal{U}}$ onto the difference-span subspace $\mathcal{U}$, and the component $\theta_{\mathcal{U}^\perp}$ is not identifiable from choice data.

Without loss of generality. Let $d' := \dim(\mathcal{U})$ and fix any orthonormal basis of $\mathcal{U}$ to represent parameters in $\mathbb{R}^{d'}$. Since (A.4) shows that replacing θ by $\theta_{\mathcal{U}}$ leaves all MNL probabilities unchanged, we may restrict the parameter space to $\mathcal{U}$ (and redefine $d \leftarrow d'$) without loss of generality. Under this reparameterization, the spanning condition in Assumption 3.1 holds by construction.

B. Supplementary Material for Section 3

B.1. Proof of Theorem 3.3

Proof of Theorem 3.3. In this subsection, we provide the proof of Theorem 3.3, showing that the LMO optimization problem in (3) reduces to a 0–1 Mixed Integer Linear Program (MILP).

Fix the current Frank–Wolfe iterate π_m and let $\mathbf{M}_m := \mathbf{M}_{\theta_0}(\pi_m) > 0$. Recall the definition of MNL choice probabilities (at θ_0):

$$p(i|S, \theta_0) := \frac{\exp(\mathbf{a}_i^\top \theta_0)}{\sum_{j \in S} \exp(\mathbf{a}_j^\top \theta_0)}, \quad \forall i \in S,$$

and the corresponding Fisher information matrix:

$$\mathbf{I}_{\theta_0}(S) := \sum_{i \in S} p(i|S, \theta_0) (\mathbf{a}_i - \bar{\mathbf{a}}_S(\theta_0)) (\mathbf{a}_i - \bar{\mathbf{a}}_S(\theta_0))^\top, \quad \text{where } \bar{\mathbf{a}}_S(\theta_0) := \sum_{j \in S} p(j|S, \theta_0) \mathbf{a}_j.$$

Then, the linear maximization oracle (LMO) selects an extreme point supported on a single assortment

$$S_m \in \operatorname{argmax}_{S \in \mathcal{S}} \operatorname{tr}(\mathbf{M}_m^{-1} \mathbf{I}_{\theta_0}(S)).$$

Binary encoding. We represent an assortment S by a binary vector $x \in \{0, 1\}^N$, where $x_i = \mathbf{1}\{i \in S\}$. This encoding ensures that $\sum_{i=1}^N x_i = |S|$, where $2 \leq |S| \leq K$. For a simple presentation, we define

$$w_i := \exp(\mathbf{a}_i^\top \theta_0) > 0, \quad w := (w_1, \dots, w_N)^\top, \quad W(x) := w^\top x = \sum_{i=1}^N w_i x_i.$$

Then, for any $i \in S$, the MNL choice probability can be written as

$$p(i|S, \theta_0) = \frac{w_i x_i}{W(x)}.$$

Moreover, the mean feature vector satisfies

$$\bar{\mathbf{a}}_S(\theta_0) = \sum_{j \in S} p(j|S, \theta_0) \mathbf{a}_j = \frac{\sum_{j=1}^N w_j x_j \mathbf{a}_j}{W(x)}.$$

Reformulation as a 0–1 QFIP. Using the definition of $\mathbf{I}_{\theta_0}(S)$, we obtain

$$\begin{aligned} \operatorname{tr}(\mathbf{M}_m^{-1} \mathbf{I}_{\theta_0}(S)) &= \sum_{i \in S} p(i|S, \theta_0) \underbrace{\mathbf{a}_i^\top \mathbf{M}_m^{-1} \mathbf{a}_i}_{=: s_i} - \bar{\mathbf{a}}_S(\theta_0)^\top \mathbf{M}_m^{-1} \bar{\mathbf{a}}_S(\theta_0) \\ &= \frac{\sum_{i=1}^N x_i \overbrace{w_i s_i}^{=: r_i}}{W(x)} - \frac{\left(\sum_{i=1}^N w_i x_i \mathbf{a}_i\right)^\top \mathbf{M}_m^{-1} \left(\sum_{j=1}^N w_j x_j \mathbf{a}_j\right)}{W(x)^2} \\ &= \frac{r^\top x}{W(x)} - \frac{\left(\sum_{i=1}^N w_i x_i \mathbf{a}_i\right)^\top \mathbf{M}_m^{-1} \left(\sum_{j=1}^N w_j x_j \mathbf{a}_j\right)}{W(x)^2}, \end{aligned} \tag{B.1}$$

where we defined

$$s_i := \mathbf{a}_i^\top \mathbf{M}_m^{-1} \mathbf{a}_i, \quad r_i := w_i s_i, \quad r := (r_1, \dots, r_N)^\top.$$

Define the symmetric matrix $G \in \mathbb{R}^{N \times N}$ by $G_{ij} := w_i w_j \mathbf{a}_i^\top \mathbf{M}_m^{-1} \mathbf{a}_j$. Then we have

$$\left(\sum_{i=1}^N w_i x_i \mathbf{a}_i\right)^\top \mathbf{M}_m^{-1} \left(\sum_{j=1}^N w_j x_j \mathbf{a}_j\right) = x^\top G x.$$

Moreover, since $W(x)^2 = (w^\top x)^2 = x^\top (w w^\top) x$, the right-hand side of (B.1) becomes

$$\operatorname{tr}(\mathbf{M}_m^{-1} \mathbf{I}_{\theta_0}(S)) = \frac{(w^\top x)(r^\top x) - x^\top G x}{(w^\top x)^2}. \tag{B.2}$$

Finally, symmetrizing the bilinear term yields a pure quadratic form:

$$(w^\top x)(r^\top x) = x^\top \left(\frac{1}{2}(w r^\top + r w^\top)\right) x.$$

Hence, by defining

$$A := -\frac{1}{2}(wr^\top + rw^\top) + G, \quad B := ww^\top,$$

we can rewrite (B.2) as

$$\text{tr}(\mathbf{M}_m^{-1} \mathbf{I}_{\theta_0}(S)) = -\frac{x^\top Ax}{x^\top Bx}.$$

Therefore, the LMO in (3) is equivalently written as the following 0–1 quadratic fractional integer problem (QFIP):

$$\max_{x \in \{0,1\}^N} -\frac{x^\top Ax}{x^\top Bx} = \min_{x \in \{0,1\}^N} \frac{x^\top Ax}{x^\top Bx}, \quad \text{s.t.} \quad 2 \leq \sum_{i=1}^N x_i \leq K. \quad (\text{B.3})$$

We next reformulate this QFIP (B.3) as a linear fractional integer program (LFIP), adapting the approach used in the proof of Theorem 2.2 of Kapoor & Arora (2006).

QFIP $\Rightarrow$ LFIP. Let $\mathbf{1} \in \mathbb{R}^N$ denote the all-ones vector and define the row-sum bounds

$$M_A := \max_{i \in [N]} \sum_{j=1}^N |A_{ij}|, \quad M_B := \max_{i \in [N]} \sum_{j=1}^N B_{ij}.$$

For any $x \in \{0,1\}^N$, we have $(Ax)_i \in [-M_A, M_A]$, hence

$$\tilde{A}x := Ax + M_A \mathbf{1} \in [0, 2M_A]^N.$$

Introduce auxiliary vectors $y^0, y^1 \in \mathbb{R}_+^N$ such that

$$\tilde{A}x = y^0 + y^1, \quad 0 \leq y^0 \leq 2M_A(\mathbf{1} - x), \quad 0 \leq y^1 \leq 2M_A x. \quad (\text{B.4})$$

These constraints enforce $y_i^1 = (\tilde{A}x)_i$ when $x_i = 1$ and $y_i^1 = 0$ when $x_i = 0$; thus

$$\mathbf{1}^\top y^1 = x^\top \tilde{A}x = x^\top Ax + M_A \mathbf{1}^\top x \implies x^\top Ax = \mathbf{1}^\top y^1 - M_A \mathbf{1}^\top x.$$

Similarly, since $B = ww^\top$ has nonnegative entries, we have $Bx \in [0, M_B]^N$. Introduce $z^0, z^1 \in \mathbb{R}_+^N$ such that

$$Bx = z^0 + z^1, \quad 0 \leq z^0 \leq M_B(\mathbf{1} - x), \quad 0 \leq z^1 \leq M_B x. \quad (\text{B.5})$$

Then $\mathbf{1}^\top z^1 = x^\top Bx$ holds, because $\mathbf{1}^\top z^1 = \sum_{i=1}^N x_i (Bx)_i = x^\top Bx$.

Therefore, (B.3) is equivalent to the following 0–1 linear fractional integer problem (LFIP):

$$\begin{aligned} \min_{x \in \{0,1\}^N, y^0, y^1, z^0, z^1 \in \mathbb{R}_+^N} \quad & \frac{\mathbf{1}^\top y^1 - M_A \mathbf{1}^\top x}{\mathbf{1}^\top z^1} \\ \text{s.t.} \quad & Ax + M_A \mathbf{1} = y^0 + y^1, \\ & 0 \leq y^0 \leq 2M_A(\mathbf{1} - x), \\ & 0 \leq y^1 \leq 2M_A x, \\ & Bx = z^0 + z^1, \\ & 0 \leq z^0 \leq M_B(\mathbf{1} - x), \\ & 0 \leq z^1 \leq M_B x, \\ & 2 \leq \mathbf{1}^\top x \leq K. \end{aligned} \quad (\text{B.6})$$

LFIP $\Rightarrow$ MILP (Charnes–Cooper). We now reduce the LFIP (B.6) to an equivalent 0–1 mixed integer linear program (MILP) via the Charnes–Cooper transformation (Cooper et al., 1962). Since $B = ww^\top$ and $w_i > 0$, the cardinality constraint $\mathbf{1}^\top x \geq 2$ implies

$$w^\top x \geq 2w_{\min}, \quad w_{\min} := \min_{i \in [N]} w_i, \quad \Rightarrow \quad x^\top Bx = (w^\top x)^2 \geq 4w_{\min}^2.$$

Hence the denominator in (B.6) satisfies $\mathbf{1}^\top z^1 = x^\top Bx \geq 4w_{\min}^2$. Let

$$\alpha := \frac{1}{\mathbf{1}^\top z^1} \in (0, \bar{\alpha}], \quad \bar{\alpha} := \frac{1}{4w_{\min}^2},$$

and define the scaled variables

$$v := \alpha x, \quad S^0 := \alpha y^0, \quad S^1 := \alpha y^1, \quad U^0 := \alpha z^0, \quad U^1 := \alpha z^1.$$

Then the objective becomes linear:

$$\frac{\mathbf{1}^\top y^1 - M_A \mathbf{1}^\top x}{\mathbf{1}^\top z^1} = \alpha(\mathbf{1}^\top y^1 - M_A \mathbf{1}^\top x) = \mathbf{1}^\top S^1 - M_A \mathbf{1}^\top v.$$

Moreover, the constraints in (B.6) transform into

$$\begin{aligned} Av + M_A \alpha \mathbf{1} &= S^0 + S^1, \\ 0 &\leq S^0 \leq 2M_A(\alpha \mathbf{1} - v), \\ 0 &\leq S^1 \leq 2M_A v, \\ Bv &= U^0 + U^1, \\ 0 &\leq U^0 \leq M_B(\alpha \mathbf{1} - v), \\ 0 &\leq U^1 \leq M_B v, \\ \mathbf{1}^\top U^1 &= 1, \\ 2 &\leq \mathbf{1}^\top x \leq K. \end{aligned}$$

Finally, we linearize the bilinear relation $v = \alpha x$ via standard big- M constraints: for all $i \in [N]$,

$$0 \leq v_i \leq \alpha, \quad v_i \leq \bar{\alpha} x_i, \quad v_i \geq \alpha - \bar{\alpha}(1 - x_i),$$

together with $x \in \{0, 1\}^N$ and $0 \leq \alpha \leq \bar{\alpha}$.

Collecting all constraints, we obtain the following equivalent 0–1 MILP:

$$\begin{aligned} \min \quad & \mathbf{1}^\top S^1 - M_A \mathbf{1}^\top v \\ \text{s.t.} \quad & Av + M_A \alpha \mathbf{1} = S^0 + S^1, \\ & 0 \leq S^0 \leq 2M_A(\alpha \mathbf{1} - v), \\ & 0 \leq S^1 \leq 2M_A v, \\ & Bv = U^0 + U^1, \\ & 0 \leq U^0 \leq M_B(\alpha \mathbf{1} - v), \\ & 0 \leq U^1 \leq M_B v, \\ & \mathbf{1}^\top U^1 = 1, \\ & 0 \leq v_i \leq \alpha, \quad \forall i \in [N], \\ & v_i \leq \bar{\alpha} x_i, \quad \forall i \in [N], \\ & v_i \geq \alpha - \bar{\alpha}(1 - x_i), \quad \forall i \in [N], \\ & 2 \leq \mathbf{1}^\top x \leq K, \\ & x \in \{0, 1\}^N, \quad 0 \leq \alpha \leq \bar{\alpha}, \\ & (v, S^0, S^1, U^0, U^1) \in \mathbb{R}_+^N \times \mathbb{R}_+^N \times \mathbb{R}_+^N \times \mathbb{R}_+^N \times \mathbb{R}_+^N. \end{aligned} \tag{B.7}$$

This completes the reduction of the LMO subproblem to a 0–1 MILP.

Consequently, since (B.3) is exactly reformulated as the 0–1 MILP (B.7), any feasible solution returned by a standard MILP solver based on techniques such as branch-and-bound and branch-and-cut, together with valid primal and dual bounds, immediately yields a corresponding solution for (B.3) with a certified suboptimality guarantee (Wolsey & Nemhauser, 1999; Wolsey, 2020). In particular, suppose the solver terminates with an incumbent feasible solution $\hat{x}$ of objective value UB (upper bound) and a provable lower bound LB (lower bound) on the optimal value, such that the certified (absolute) optimality gap satisfies

$$\text{UB} - \text{LB} \leq \epsilon_{\text{LMO}}.$$

Let x^* be an optimal solution to (B.3). By exact equivalence of the MILP and (B.3), the MILP optimum equals the optimal value of (B.3), and therefore

$$\frac{(\hat{x})^\top A \hat{x}}{(\hat{x})^\top B \hat{x}} - \frac{(x^*)^\top A x^*}{(x^*)^\top B x^*} \leq \text{UB} - \text{LB} \leq \epsilon_{\text{LMO}}.$$

Hence, solving the MILP to a certified optimality gap of at most ϵ_{LMO} yields an ϵ_{LMO} -optimal solution to the LMO problem (B.3) (equivalently, (3)). This completes the proof of Theorem 3.3. $\square$

B.2. Tightening big- M constants in (B.6)–(B.7)

The proof in Appendix B.1 provides an *exact* reformulation of the LMO subproblem as a 0–1 MILP, using the bounds M_A , M_B , and $\bar{\alpha}$, which are usually referred to as big- M constants (Wolsey & Nemhauser, 1999; Wolsey, 2020). In practice, however, the runtime of MILP solvers—through the strength of the LP relaxation and the effectiveness of branching—is often highly sensitive to how tight these bounds are. Motivated by this, we derive tighter, problem-specific bounds that retain exact equivalence while typically leading to substantial computational speedups.

Throughout, we only require bounds that are valid over the feasible set $\{x \in \{0, 1\}^N : 2 \leq \mathbf{1}^\top x \leq K\}$. Replacing $(M_A, M_B, \bar{\alpha})$ by any *smaller* constants/vectors that remain valid upper bounds does not remove any feasible (x, y^0, y^1, z^0, z^1) that corresponds to a feasible x of the original QFIP, and therefore preserves the exact equivalence between (B.3), (B.6), and (B.7). We now describe concrete tighter choices.

B.2.1. TIGHTER BOUNDS FOR $(Ax)_i$ EXPLOITING $\mathbf{1}^\top x \leq K$

The coarse bound $M_A := \max_i \sum_j |A_{ij}|$ ignores sparsity. Since any feasible x has at most K ones, we can tighten the bound on each coordinate $(Ax)_i$.

For each row $i \in [N]$, let $\{|A_{ij}|\}_{j=1}^N$ be sorted in nonincreasing order:

$$|A_{i(1)}| \geq |A_{i(2)}| \geq \dots \geq |A_{i(N)}|.$$

Define the K -aware row bound

$$M_{A,i}^{(K)} := \sum_{\ell=1}^K |A_{i(\ell)}| \quad \text{and} \quad M_A^{(K)} := \max_{i \in [N]} M_{A,i}^{(K)}.$$

Then for any feasible x with $\mathbf{1}^\top x \leq K$,

$$|(Ax)_i| = \left| \sum_{j=1}^N A_{ij} x_j \right| \leq \sum_{j: x_j=1} |A_{ij}| \leq \sum_{\ell=1}^K |A_{i(\ell)}| = M_{A,i}^{(K)} \leq M_A^{(K)}.$$

Option 1: minimal-change replacement (scalar bound). One may simply replace M_A by $M_A^{(K)}$ in (B.4) and everywhere thereafter:

$$\tilde{A}x := Ax + M_A^{(K)} \mathbf{1} \in [0, 2M_A^{(K)}]^N, \quad 0 \leq y^0 \leq 2M_A^{(K)}(\mathbf{1} - x), \quad 0 \leq y^1 \leq 2M_A^{(K)}x,$$

and the identity becomes

$$x^\top Ax = \mathbf{1}^\top y^1 - M_A^{(K)} \mathbf{1}^\top x.$$

All subsequent steps remain unchanged with this substitution.

Option 2: Stronger replacement (row-wise bound). A further tightening uses row-dependent shifts. Let $m_A \in \mathbb{R}_+^N$ be given by $(m_A)_i := M_{A,i}^{(K)}$ and define

$$\tilde{A}x := Ax + m_A \in [0, 2m_A] \quad (\text{componentwise}).$$

Then replace (B.4) by

$$\tilde{A}x = y^0 + y^1, \quad 0 \leq y^0 \leq 2m_A \odot (\mathbf{1} - x), \quad 0 \leq y^1 \leq 2m_A \odot x, \quad (\text{B.8})$$

where $\odot$ denotes elementwise product. The same logic yields

$$\mathbf{1}^\top y^1 = \sum_{i=1}^N x_i (\tilde{A}x)_i = x^\top (Ax + m_A) = x^\top Ax + m_A^\top x, \implies x^\top Ax = \mathbf{1}^\top y^1 - m_A^\top x.$$

Consequently, the LFIP objective in (B.6) becomes

$$\frac{\mathbf{1}^\top y^1 - m_A^\top x}{\mathbf{1}^\top z^1} \quad (\text{instead of } \frac{\mathbf{1}^\top y^1 - M_A \mathbf{1}^\top x}{\mathbf{1}^\top z^1}).$$

All other constraints remain identical after replacing M_A by m_A componentwise in the upper bounds.

B.2.2. TIGHTER BOUNDS FOR $(Bx)_i$ EXPLOITING $B = ww^\top$ AND $\mathbf{1}^\top x \leq K$

The coarse bound $M_B := \max_i \sum_j B_{ij}$ is also loose. Here we can exploit the rank-one structure $B = ww^\top$:

$$(Bx)_i = \sum_{j=1}^N w_i w_j x_j = w_i \underbrace{\left(\sum_{j=1}^N w_j x_j \right)}_{=: W(x)} = w_i W(x).$$

Let $w_{[1]}^\downarrow \geq \dots \geq w_{[N]}^\downarrow$ be the weights sorted in decreasing order and define

$$W_{\max}^{(K)} := \sum_{\ell=1}^K w_{[\ell]}^\downarrow.$$

Then for any feasible x with $\mathbf{1}^\top x \leq K$,

$$0 \leq W(x) \leq W_{\max}^{(K)} \implies 0 \leq (Bx)_i = w_i W(x) \leq w_i W_{\max}^{(K)}.$$

Define the componentwise bound vector $m_B \in \mathbb{R}_+^N$ by

$$(m_B)_i := w_i W_{\max}^{(K)}.$$

Then one may replace (B.5) by the tighter, componentwise constraints

$$Bx = z^0 + z^1, \quad 0 \leq z^0 \leq m_B \odot (\mathbf{1} - x), \quad 0 \leq z^1 \leq m_B \odot x. \quad (\text{B.9})$$

Exactly as before, these enforce $\mathbf{1}^\top z^1 = x^\top Bx$. The same componentwise replacements propagate through the Charnes–Cooper step: after scaling $(U^0, U^1) = \alpha(z^0, z^1)$ and $v = \alpha x$, we obtain

$$0 \leq U^0 \leq m_B \odot (\alpha \mathbf{1} - v), \quad 0 \leq U^1 \leq m_B \odot v$$

in place of the coarse bounds with M_B .

B.2.3. TIGHTER UPPER BOUND ON α EXPLOITING $\mathbf{1}^\top x \geq 2$

In the proof above we used $w^\top x \geq 2w_{\min}$ to deduce $\bar{\alpha} = 1/(4w_{\min}^2)$. This can be tightened by using the exact minimal possible sum of two weights.

Let $w_{(1)}^\uparrow \leq w_{(2)}^\uparrow \leq \dots \leq w_{(N)}^\uparrow$ be the weights sorted in increasing order and define

$$W_{\min}^{(2)} := w_{(1)}^\uparrow + w_{(2)}^\uparrow.$$

Then the constraint $\mathbf{1}^\top x \geq 2$ implies $w^\top x \geq W_{\min}^{(2)}$, and therefore

$$x^\top Bx = (w^\top x)^2 \geq (W_{\min}^{(2)})^2, \quad \Rightarrow \quad \alpha = \frac{1}{\mathbf{1}^\top z \mathbf{1}} = \frac{1}{x^\top Bx} \leq \frac{1}{(W_{\min}^{(2)})^2}.$$

Hence one may replace the coarse upper bound $\bar{\alpha} = 1/(4w_{\min}^2)$ by the tighter

$$\bar{\alpha}_{\text{tight}} := \frac{1}{(W_{\min}^{(2)})^2}.$$

This directly strengthens the big- M linearization of $v = \alpha x$:

$$v_i \leq \bar{\alpha}_{\text{tight}} x_i, \quad v_i \geq \alpha - \bar{\alpha}_{\text{tight}}(1 - x_i), \quad \forall i \in [N].$$

More generally, if the feasible set enforces a different lower cardinality $|S| \geq L_{\min}$, one can use $W_{\min}^{(L_{\min})} := \sum_{\ell=1}^{L_{\min}} w_{(\ell)}^\uparrow$ and obtain $\bar{\alpha}_{\text{tight}} = 1/(W_{\min}^{(L_{\min})})^2$.

B.3. Proof of Proposition 3.2

Proof of Proposition 3.2. Fix $\theta_0 \in \mathbb{R}^d$. We apply the following Lemma B.1 by indexing $[L]$ with assortments $S \in \mathcal{S}$ and choosing, for each $S \in \mathcal{S}$, a matrix $\mathbf{A}_S$ satisfying $\mathbf{I}_{\theta_0}(S) = \mathbf{A}_S \mathbf{A}_S^\top$ (e.g., any matrix square-root factor).

Lemma B.1 (Theorem 1 of Mukherjee et al. 2024). *Let $M \geq 1$ be an integer and $\mathbf{A}_1, \dots, \mathbf{A}_L \in \mathbb{R}^{d \times M}$ be L matrices whose column space spans $\mathbb{R}^d$. For any probability distribution $\pi \in \Delta_L$ over $[L] := \{1, \dots, L\}$, define the objective*

$$g(\pi) := \max_{i \in [L]} \text{tr}(\mathbf{A}_i^\top \mathbf{V}_\pi^{-1} \mathbf{A}_i),$$

where $\mathbf{V}_\pi := \sum_{i=1}^L \pi_i \mathbf{A}_i \mathbf{A}_i^\top$ is a design matrix. Then, the following statements are equivalent:

- (a) π^* is a minimizer of $g(\pi)$.
- (b) π^* is a maximizer of $f(\pi) = \log \det(\mathbf{V}_\pi)$.
- (c) $g(\pi^*) = d$.

Furthermore, there exists a minimizer π^* of $g(\pi)$ such that $|\text{supp}(\pi^*)| \leq d(d+1)/2$.

Under Assumption 3.1, the collection of columns of $\{\mathbf{A}_S\}_{S \in \mathcal{S}}$ spans $\mathbb{R}^d$, so the spanning condition in Lemma B.1 is satisfied. For any design π over $\mathcal{S}$, the induced design matrix in Lemma B.1 becomes

$$\mathbf{V}_\pi = \mathbb{E}_{S \sim \pi} [\mathbf{A}_S \mathbf{A}_S^\top] = \mathbb{E}_{S \sim \pi} [\mathbf{I}_{\theta_0}(S)] = \mathbf{M}_{\theta_0}(\pi).$$

Moreover, for each $S \in \mathcal{S}$,

$$\text{tr}(\mathbf{A}_S^\top \mathbf{V}_\pi^{-1} \mathbf{A}_S) = \text{tr}(\mathbf{V}_\pi^{-1} \mathbf{A}_S \mathbf{A}_S^\top) = \text{tr}(\mathbf{M}_{\theta_0}(\pi)^{-1} \mathbf{I}_{\theta_0}(S)),$$

so the objective $g(\pi)$ in Lemma B.1 coincides with $g_{\theta_0}(\pi)$ in (2). Similarly, the D-optimal objective $f(\pi) = \log \det(\mathbf{V}_\pi)$ coincides with $f_{\theta_0}(\pi) = \log \det(\mathbf{M}_{\theta_0}(\pi))$. Therefore, applying Lemma B.1 yields the equivalence of (a)–(c) in Proposition 3.2.

Finally, the support-size guarantee follows directly from the last statement of Lemma B.1, implying that there exists a minimizer $\pi_{\theta_0}^*$ of $g_{\theta_0}(\pi)$ such that $|\text{supp}(\pi_{\theta_0}^*)| \leq d(d+1)/2$. $\square$

Algorithm B.1 Frank–Wolfe Algorithm for MNL at θ_0 with ϵ_{LMO} -approximate LMO

- 1: **Input:** nominal parameter θ_0 , initial distribution $\pi_0 \in \Delta(\mathcal{S})$, FW precision ε , LMO certified gap tolerance ϵ_{LMO} .
- 2: **Initialize:** $\tilde{\varepsilon} \leftarrow \varepsilon - \epsilon_{\text{LMO}}/d > 0$, $\mathbf{M}_0 \leftarrow \mathbf{M}_{\theta_0}(\pi_0) > 0$, $m \leftarrow 0$.
- 3: **while** $g_{\theta_0}(\pi_m) > (1 + \tilde{\varepsilon})d$ **do**
- 4: Run a MILP solver for $\hat{S}_m \leftarrow \max_{S \in \mathcal{S}} \text{tr}(\mathbf{M}_m^{-1} \mathbf{I}_{\theta_0}(S))$ with certified gap $\leq \epsilon_{\text{LMO}}$.
- 5: $\gamma_m \leftarrow \arg\max_{\gamma \in [0,1]} f_{\theta_0}((1 - \gamma)\pi_m + \gamma \mathbb{1}_{\hat{S}_m})$.
- 6: $\pi_{m+1} \leftarrow (1 - \gamma_m)\pi_m + \gamma_m \mathbb{1}_{\hat{S}_m}$.
- 7: Update $\mathbf{M}_{m+1} \leftarrow \mathbf{M}_{\theta_0}(\pi_{m+1})$ and set $m \leftarrow m + 1$.
- 8: **end while**
- 9: Output $\hat{\pi}_{\theta_0} \leftarrow \pi_m$.

B.4. Proof of Proposition 3.4

Proposition B.2 (FW stopping-time guarantee with an approximate LMO, extended version of Proposition 3.4). *Fix θ_0 . Let π_m be the Frank–Wolfe iterate and write $\mathbf{M}_m := \mathbf{M}_{\theta_0}(\pi_m)$. Define the certificate $g_m^* := \max_{S \in \mathcal{S}} \text{tr}(\mathbf{M}_m^{-1} \mathbf{I}_{\theta_0}(S))$, and $\hat{g}_m := \text{tr}(\mathbf{M}_m^{-1} \mathbf{I}_{\theta_0}(\hat{S}_m))$, where $\hat{S}_m \in \mathcal{S}$ is returned by an approximate LMO at iteration m . Assume the per-iteration LMO guarantee $g_m^* - \hat{g}_m \leq \epsilon_{\text{LMO}}$. Define the stopping time based on the computable certificate*

$$\tau := \inf \left\{ m \geq 0 : \hat{g}_m \leq (1 + \tilde{\varepsilon})d \right\}, \quad \text{where } \tilde{\varepsilon} := \varepsilon - \frac{\epsilon_{\text{LMO}}}{d}. \quad (\text{B.10})$$

Given ϵ_{LMO} , set ϵ such that $1 \geq \tilde{\varepsilon} = \varepsilon - \epsilon_{\text{LMO}}/d > 0$. Suppose that, throughout the run up to the stopping time, the iterates satisfy $\mathbf{M}_m \geq \lambda_0 \mathbf{I}_d$ for some constant $\lambda_0 > 0$, and that $\|\mathbf{I}_{\theta_0}(S)\|_2 \leq L$ for all S . Then, we have

- (i) **Certified accuracy at the stopping time.** At termination, the returned iterate π_τ satisfies $g_\tau^* \leq (1 + \epsilon)d$.
- (ii) **Finite stopping and an iteration bound.** The algorithm terminates in finite time, and the stopping time satisfies

$$\tau \leq \mathcal{O}\left(\frac{d \log(1/\lambda_0)}{\tilde{\varepsilon}}\right).$$

Proof of Proposition B.2. The first claim follows immediately from the stopping rule. By definition of the stopping time τ in (B.10), we have $\hat{g}_\tau \leq (1 + \tilde{\varepsilon})d = (1 + \epsilon)d - \epsilon_{\text{LMO}}$. Moreover, by the assumed LMO accuracy $g_m^* - \hat{g}_m \leq \epsilon_{\text{LMO}}$ for all m , it follows that

$$g_\tau^* \leq \hat{g}_\tau + \epsilon_{\text{LMO}} \leq (1 + \epsilon)d,$$

which establishes claim (i).

We now prove claim (ii). Fix any m and write $\mathbf{I}_m := \mathbf{I}_{\theta_0}(\hat{S}_m)$. For any $\gamma \in [0, 1]$, define

$$\pi_m(\gamma) := (1 - \gamma)\pi_m + \gamma \mathbf{1}_{\hat{S}_m}, \quad \mathbf{M}_m(\gamma) := \mathbf{M}_{\theta_0}(\pi_m(\gamma)) = (1 - \gamma)\mathbf{M}_m + \gamma \mathbf{I}_m.$$

Then, we have

$$\begin{aligned} f(\pi_m(\gamma)) - f(\pi_m) &= \log \det(\mathbf{M}_m(\gamma)) - \log \det(\mathbf{M}_m) \\ &= \log \det\left((1 - \gamma)\mathbf{I}_d + \gamma \mathbf{M}_m^{-1/2} \mathbf{I}_m \mathbf{M}_m^{-1/2}\right). \end{aligned}$$

Let $\mathbf{L}_m := \mathbf{M}_m^{-1/2} \mathbf{I}_m \mathbf{M}_m^{-1/2} \geq 0$ and denote its eigenvalues by $\{\lambda_i\}_{i=1}^d$. Then, by the AM–GM inequality, we get

$$f(\pi_m(\gamma)) - f(\pi_m) = \det\left((1 - \gamma)\mathbf{I}_d + \gamma \mathbf{L}_m\right) = \prod_{i=1}^d ((1 - \gamma) + \gamma \lambda_i) \geq \left((1 - \gamma) + \gamma \cdot \frac{1}{d} \sum_{i=1}^d \lambda_i\right)^d.$$

Noting that $\sum_{i=1}^d \lambda_i = \text{tr}(\mathbf{L}_m) = \text{tr}(\mathbf{M}_m^{-1} \mathbf{I}_m) = \hat{g}_m$, we obtain

$$f(\pi_{m+1}) = \max_{\gamma' \in [0,1]} f(\pi_{m+1}(\gamma')) \geq f(\pi_m(\gamma)) - f(\pi_m) \geq d \log\left((1 - \gamma) + \gamma \frac{\hat{g}_m}{d}\right). \quad (\text{B.11})$$

Choose $\gamma_m := 1/\hat{g}_m$. Let τ be the stopping time. For $m < \tau$, we have $\hat{g}_m > (1 + \tilde{\epsilon})d > d$, hence $\gamma_m \in (0, 1)$. Plugging $\gamma = \gamma_m$ into (B.11) yields

$$f(\pi_{m+1}) - f(\pi_m) \geq d \log \left(1 - \frac{1}{\hat{g}_m} + \frac{1}{d} \right) = d \log \left(1 + \frac{1}{d} - \frac{1}{\hat{g}_m} \right).$$

If $m < \tau$, then $\hat{g}_m > (1 + \tilde{\epsilon})d$, hence

$$\frac{1}{d} - \frac{1}{\hat{g}_m} \geq \frac{1}{d} - \frac{1}{(1 + \tilde{\epsilon})d} = \frac{\tilde{\epsilon}}{(1 + \tilde{\epsilon})d}.$$

Therefore, we obtain

$$f(\pi_{m+1}) - f(\pi_m) \geq d \log \left(1 + \frac{\tilde{\epsilon}}{(1 + \tilde{\epsilon})d} \right). \quad (\text{B.12})$$

Let $x := \frac{\tilde{\epsilon}}{(1 + \tilde{\epsilon})d}$. Since $d \geq 1$ and $\frac{\tilde{\epsilon}}{1 + \tilde{\epsilon}} \leq 1$, we have $x \in (0, 1]$. Using $\log(1 + x) \geq x/2$ for $x \in [0, 1]$, from (B.12) we get

$$f(\pi_{m+1}) - f(\pi_m) \geq d \log(1 + x) \geq d \cdot \frac{x}{2} = \frac{\tilde{\epsilon}}{2(1 + \tilde{\epsilon})}.$$

If additionally $\tilde{\epsilon} \leq 1$, then $\frac{1}{1 + \tilde{\epsilon}} \geq \frac{1}{2}$ and hence

$$f(\pi_{m+1}) - f(\pi_m) \geq \frac{\tilde{\epsilon}}{4}. \quad (\text{B.13})$$

By assumption, for any $m \leq \tau$, $\mathbf{M}_m \geq \lambda_0 \mathbf{I}_d$, so $f(\pi_m) = \log \det(\mathbf{M}_m) \geq d \log \lambda_0$. Moreover, since $\mathbf{I}_{\theta_0}(S) \geq 0$ and $\|\mathbf{I}_{\theta_0}(S)\|_2 \leq L$, we have $\mathbf{I}_{\theta_0}(S) \leq L \mathbf{I}_d$ for all S . Taking convex combinations yields $\mathbf{M}_m \leq L \mathbf{I}_d$, hence $f(\pi_m) \leq d \log L$. Therefore,

$$f(\pi_m) - f(\pi_0) \leq d \log(L/\lambda_0) \quad \forall m \leq \tau.$$

On the other hand, by (B.13), each iteration prior to stopping increases f by at least $\tilde{\epsilon}/4$: for any integer $T < \tau$,

$$f(\pi_T) - f(\pi_0) = \sum_{m=0}^{T-1} (f(\pi_{m+1}) - f(\pi_m)) \geq T \cdot \frac{\tilde{\epsilon}}{4}.$$

Combining gives

$$T \cdot \frac{\tilde{\epsilon}}{4} \leq d \log(L/\lambda_0), \quad \forall T < \tau,$$

so τ is finite and satisfies

$$\tau \leq 1 + \left\lceil \frac{4d \log(L/\lambda_0)}{\tilde{\epsilon}} \right\rceil.$$

This proves claim (ii). □

B.5. Proof of Theorem 3.5

Theorem B.3 (Lifted FW guarantee, restatement of Theorem 3.5). *Fix θ_0 and run Frank–Wolfe (Algorithm 1) to minimize $\tilde{g}_{\theta_0}(\pi)$ using the polynomial-time lifted LMO (6). Let $\hat{\pi}$ be the design returned at the stopping time $\tilde{g}_{\theta_0}(\hat{\pi}) \leq (1 + \epsilon)(d + 1)$, and define*

$$\epsilon_{\text{lift}} := \inf \{ \epsilon \geq 0 : \Delta_{\theta_0}(\hat{\pi}) \leq \epsilon \mathbf{M}_{\theta_0}(\hat{\pi}) \}.$$

Then

$$g_{\theta_0}(\hat{\pi}) \leq (1 + \epsilon_{\text{lift}})((1 + \epsilon)d + \epsilon) \leq 2(1 + \epsilon_{\text{lift}})(1 + \epsilon)d.$$

Proof of Theorem 3.5. Throughout the proof, fix an arbitrary design $\pi \in \Delta(\mathcal{S})$ and abbreviate

$$\mathbf{M} := \mathbf{M}_{\theta_0}(\pi) \in \mathbb{R}^{d \times d}, \quad \widetilde{\mathbf{M}} := \widetilde{\mathbf{M}}_{\theta_0}(\pi) \in \mathbb{R}^{(d+1) \times (d+1)}, \quad \tilde{g} := \tilde{g}_{\theta_0}(\pi), \quad g := g_{\theta_0}(\pi).$$

Algorithm B.2 Frank–Wolfe Algorithm for *Lifted* G -optimal Design (MNL) at θ_0

- 1: **Input:** nominal parameter θ_0 , initial distribution $\pi_0 \in \Delta(S)$, FW precision ε .
- 2: **Initialize:** $\tilde{\mathbf{M}}_0 = \tilde{\mathbf{M}}_{\theta_0}(\pi_0) > 0$, $m = 0$.
- 3: **Define:** $\tilde{f}_{\theta_0}(\pi) := \log \det(\tilde{\mathbf{M}}_{\theta_0}(\pi))$.
- 4: **while** $\tilde{g}_{\theta_0}(\pi_m) > (1 + \varepsilon)(d + 1)$ **do**
- 5: $S_m \leftarrow \operatorname{argmax}_{S \in \mathcal{S}} \operatorname{tr}(\tilde{\mathbf{M}}_m^{-1} \tilde{\mathbf{I}}_{\theta_0}(S))$.
- 6: $\gamma_m \leftarrow \operatorname{argmax}_{\gamma \in [0,1]} \tilde{f}_{\theta_0}((1 - \gamma)\pi_m + \gamma \mathbb{1}_{S_m})$.
- 7: $\pi_{m+1} = (1 - \gamma_m)\pi_m + \gamma_m \mathbb{1}_{S_m}$.
- 8: Update $\tilde{\mathbf{M}}_{m+1} = \tilde{\mathbf{M}}_{\theta_0}(\pi_{m+1})$ and $m \leftarrow m + 1$.
- 9: **end while**
- 10: Output the estimated lifted design $\tilde{\pi}_{\theta_0}^* = \pi_m$.

Write the block decomposition of $\tilde{\mathbf{M}}$ as

$$\tilde{\mathbf{M}} = \begin{bmatrix} \bar{\mathbf{A}} & \bar{\mathbf{b}} \\ \bar{\mathbf{b}}^\top & 1 \end{bmatrix}, \quad \bar{\mathbf{A}} := \mathbb{E}_{S \sim \pi} [\bar{\mathbf{A}}_{\theta_0}(S)], \quad \bar{\mathbf{b}} := \mathbb{E}_{S \sim \pi} [\bar{\mathbf{a}}_{\theta_0}(S)].$$

Define the Schur-complement matrix

$$\widehat{\mathbf{M}} := \bar{\mathbf{A}} - \bar{\mathbf{b}}\bar{\mathbf{b}}^\top \in \mathbb{R}^{d \times d},$$

and recall the mismatch matrix

$$\Delta_{\theta_0}(\pi) := \widehat{\mathbf{M}} - \mathbf{M} \geq 0.$$

We note that $\Delta_{\theta_0}(\pi) \geq 0$ holds directly. Indeed, recall

$$\Delta_{\theta_0}(\pi) = \widehat{\mathbf{M}} - \mathbf{M} = \bar{\mathbf{A}} - \bar{\mathbf{b}}\bar{\mathbf{b}}^\top - \left(\bar{\mathbf{A}} - \mathbb{E}_{S \sim \pi} [\bar{\mathbf{a}}_{\theta_0}(S)\bar{\mathbf{a}}_{\theta_0}(S)^\top] \right) = \mathbb{E}_{S \sim \pi} [\bar{\mathbf{a}}_{\theta_0}(S)\bar{\mathbf{a}}_{\theta_0}(S)^\top] - \bar{\mathbf{b}}\bar{\mathbf{b}}^\top,$$

For any $u \in \mathbb{R}^d$, we have

$$u^\top \Delta_{\theta_0}(\pi) u = \mathbb{E}_{S \sim \pi} [(u^\top \bar{\mathbf{a}}_{\theta_0}(S))^2] - \left(\mathbb{E}_{S \sim \pi} [u^\top \bar{\mathbf{a}}_{\theta_0}(S)] \right)^2 = \operatorname{Var}_{S \sim \pi} (u^\top \bar{\mathbf{a}}_{\theta_0}(S)) \geq 0.$$

Since u is arbitrary, this implies $\Delta_{\theta_0}(\pi) \geq 0$.

Fix any $S \in \mathcal{S}$ and abbreviate

$$\mathbf{I}(S) := \mathbf{I}_{\theta_0}(S), \quad \tilde{\mathbf{I}}(S) := \tilde{\mathbf{I}}_{\theta_0}(S).$$

From (5), we can write

$$\tilde{\mathbf{I}}(S) = \begin{bmatrix} \bar{\mathbf{A}}_{\theta_0}(S) & \bar{\mathbf{a}}_{\theta_0}(S) \\ \bar{\mathbf{a}}_{\theta_0}(S)^\top & 1 \end{bmatrix}, \quad \mathbf{I}(S) = \bar{\mathbf{A}}_{\theta_0}(S) - \bar{\mathbf{a}}_{\theta_0}(S)\bar{\mathbf{a}}_{\theta_0}(S)^\top.$$

Since the bottom-right block of $\tilde{\mathbf{M}}$ equals 1, the standard block inverse formula yields

$$\tilde{\mathbf{M}}^{-1} = \begin{bmatrix} \widehat{\mathbf{M}}^{-1} & -\widehat{\mathbf{M}}^{-1}\bar{\mathbf{b}} \\ -\bar{\mathbf{b}}^\top \widehat{\mathbf{M}}^{-1} & 1 + \bar{\mathbf{b}}^\top \widehat{\mathbf{M}}^{-1}\bar{\mathbf{b}} \end{bmatrix}. \quad (\text{B.14})$$

Now compute $\operatorname{tr}(\tilde{\mathbf{M}}^{-1}\tilde{\mathbf{I}}(S))$ using (B.14). Multiplying the blocks and taking trace gives

$$\begin{aligned} \operatorname{tr}(\tilde{\mathbf{M}}^{-1}\tilde{\mathbf{I}}(S)) &= \operatorname{tr}(\widehat{\mathbf{M}}^{-1}\bar{\mathbf{A}}_{\theta_0}(S)) - 2\bar{\mathbf{b}}^\top \widehat{\mathbf{M}}^{-1}\bar{\mathbf{a}}_{\theta_0}(S) + \left(1 + \bar{\mathbf{b}}^\top \widehat{\mathbf{M}}^{-1}\bar{\mathbf{b}}\right) \cdot 1 \\ &= 1 + \operatorname{tr}(\widehat{\mathbf{M}}^{-1}\bar{\mathbf{A}}_{\theta_0}(S)) - 2\bar{\mathbf{b}}^\top \widehat{\mathbf{M}}^{-1}\bar{\mathbf{a}}_{\theta_0}(S) + \bar{\mathbf{b}}^\top \widehat{\mathbf{M}}^{-1}\bar{\mathbf{b}}. \end{aligned}$$

Add and subtract $\bar{\mathbf{a}}_{\theta_0}(S)^\top \widehat{\mathbf{M}}^{-1} \bar{\mathbf{a}}_{\theta_0}(S)$ to complete the square:

$$\begin{aligned} \text{tr}(\widehat{\mathbf{M}}^{-1} \tilde{\mathbf{I}}(S)) &= 1 + \left[\text{tr}(\widehat{\mathbf{M}}^{-1} \bar{\mathbf{A}}_{\theta_0}(S)) - \bar{\mathbf{a}}_{\theta_0}(S)^\top \widehat{\mathbf{M}}^{-1} \bar{\mathbf{a}}_{\theta_0}(S) \right] + \|\bar{\mathbf{a}}_{\theta_0}(S) - \bar{\mathbf{b}}\|_{\widehat{\mathbf{M}}^{-1}}^2 \\ &= 1 + \text{tr}(\widehat{\mathbf{M}}^{-1} \mathbf{I}(S)) + \|\bar{\mathbf{a}}_{\theta_0}(S) - \bar{\mathbf{b}}\|_{\widehat{\mathbf{M}}^{-1}}^2. \end{aligned}$$

Therefore, for every $S \in \mathcal{S}$,

$$\text{tr}(\widehat{\mathbf{M}}^{-1} \tilde{\mathbf{I}}(S)) - 1 \geq \text{tr}(\widehat{\mathbf{M}}^{-1} \mathbf{I}(S)), \quad (\text{B.15})$$

since the norm term is nonnegative.

On the other hand, we define

$$\hat{g}_{\theta_0}(\pi) := \max_{S \in \mathcal{S}} \text{tr}(\widehat{\mathbf{M}}^{-1} \mathbf{I}(S)).$$

Then, taking maxima over S in (B.15) yields

$$\tilde{g}_{\theta_0}(\pi) - 1 \geq \hat{g}_{\theta_0}(\pi). \quad (\text{B.16})$$

Furthermore, by definition, $\widehat{\mathbf{M}} = \mathbf{M} + \Delta_{\theta_0}(\pi)$ with $\Delta_{\theta_0}(\pi) \geq 0$, so $\widehat{\mathbf{M}} \geq \mathbf{M}$ and hence $\widehat{\mathbf{M}}^{-1} \leq \mathbf{M}^{-1}$. Moreover, if $\Delta_{\theta_0}(\pi) \leq \varepsilon \mathbf{M}$ for some $\varepsilon \geq 0$, then

$$\widehat{\mathbf{M}} = \mathbf{M} + \Delta_{\theta_0}(\pi) \leq (1 + \varepsilon) \mathbf{M} \implies \mathbf{M}^{-1} \leq (1 + \varepsilon) \widehat{\mathbf{M}}^{-1}.$$

Consequently, for every $S \in \mathcal{S}$, we get

$$\text{tr}(\mathbf{M}^{-1} \mathbf{I}(S)) \leq (1 + \varepsilon) \text{tr}(\widehat{\mathbf{M}}^{-1} \mathbf{I}(S)).$$

Taking maxima over S gives the sandwich

$$g_{\theta_0}(\pi) \leq (1 + \varepsilon) \hat{g}_{\theta_0}(\pi). \quad (\text{B.17})$$

Hence, combining (B.16) and (B.17) yields: if $\Delta_{\theta_0}(\pi) \leq \varepsilon \mathbf{M}_{\theta_0}(\pi)$, then

$$g_{\theta_0}(\pi) \leq (1 + \varepsilon) (\tilde{g}_{\theta_0}(\pi) - 1). \quad (\text{B.18})$$

Now set $\pi = \hat{\pi}$ and let $\varepsilon = \epsilon_{\text{lift}}$, where by definition $\Delta_{\theta_0}(\hat{\pi}) \leq \epsilon_{\text{lift}} \mathbf{M}_{\theta_0}(\hat{\pi})$. Applying (B.18) gives

$$g_{\theta_0}(\hat{\pi}) \leq (1 + \epsilon_{\text{lift}}) (\tilde{g}_{\theta_0}(\hat{\pi}) - 1).$$

Finally, the stopping rule $\tilde{g}_{\theta_0}(\hat{\pi}) \leq (1 + \epsilon)(d + 1)$ implies

$$\tilde{g}_{\theta_0}(\hat{\pi}) - 1 \leq (1 + \epsilon)(d + 1) - 1 = (1 + \epsilon)d + \epsilon,$$

and hence

$$g_{\theta_0}(\hat{\pi}) \leq (1 + \epsilon_{\text{lift}}) ((1 + \epsilon)d + \epsilon) \leq 2(1 + \epsilon_{\text{lift}})(1 + \epsilon)d,$$

which complete the proof of Theorem 3.5. $\square$

B.6. Crude Bound for Lifted FW Error

Proposition B.4 (Crude bound for ϵ_{lift}). *If $\mathbf{M}_{\theta_0}(\hat{\pi}) \geq \lambda_0 \mathbf{I}_d$ for some $\lambda_0 > 0$, then*

$$\epsilon_{\text{lift}} \|\Delta_{\theta_0}(\hat{\pi})\|_{\text{op}} \leq \frac{1}{\lambda_0}.$$

Proof of Proposition B.4. The proof is direct:

$$\epsilon_{\text{lift}} \leq \|\mathbf{M}_{\theta_0}(\hat{\pi})^{-1}\|_{\text{op}} \|\Delta_{\theta_0}(\hat{\pi})\|_{\text{op}} \leq \frac{1}{\lambda_0} \|\Delta_{\theta_0}(\hat{\pi})\|_{\text{op}} \leq \frac{1}{\lambda_0},$$

which concludes the proof. $\square$

Proposition B.5 (Crude bound for ϵ_{lift} under the outside-option MNL model). *Under the outside-option MNL model at θ_0 , the lifting error at the returned design $\hat{\pi}$ satisfies*

$$\epsilon_{\text{lift}} \leq \sup_{S \sim \hat{\pi}} \frac{1 - p_0(S)}{p_0(S)} = \sup_{S \sim \hat{\pi}} \sum_{j \in S} \exp(\mathbf{a}_j^\top \theta_0).$$

In particular, if $|S| \leq K$ for all $S \in \mathcal{S}$ and $\max_{j \in [N]} \mathbf{a}_j^\top \theta_0 \leq B$, then

$$\epsilon_{\text{lift}} \leq K e^B.$$

Proof of Proposition B.5. For any $S \in \mathcal{S}$, define $w_i := \exp(\mathbf{a}_i^\top \theta_0)$ and the outside-option probabilities

$$p_0(S) := \frac{1}{1 + \sum_{j \in S} w_j}, \quad p_i(S) := \frac{w_i}{1 + \sum_{j \in S} w_j} \quad (i \in S).$$

Let

$$\bar{\mathbf{a}}(S) := \sum_{i \in S} p_i(S) \mathbf{a}_i, \quad \bar{\mathbf{A}}(S) := \sum_{i \in S} p_i(S) \mathbf{a}_i \mathbf{a}_i^\top, \quad \mathbf{I}(S) := \bar{\mathbf{A}}(S) - \bar{\mathbf{a}}(S) \bar{\mathbf{a}}(S)^\top \geq 0.$$

Moreover, we define

$$\mathbf{M}(\hat{\pi}) := \mathbb{E}_{S \sim \hat{\pi}} [\mathbf{I}(S)], \quad \Delta(\hat{\pi}) := \mathbb{E}_{S \sim \pi} [\bar{\mathbf{a}}(S) \bar{\mathbf{a}}(S)^\top] - \mathbb{E}_{S \sim \pi} [\bar{\mathbf{a}}(S)] \mathbb{E}_{S \sim \pi} [\bar{\mathbf{a}}(S)]^\top \geq 0,$$

and $\epsilon_{\text{lift}} := \inf \{ \varepsilon \geq 0 : \Delta(\hat{\pi}) \leq \varepsilon \mathbf{M}(\hat{\pi}) \}$. Then, by the PSD Cauchy–Schwarz inequality,

$$\bar{\mathbf{a}}(S) \bar{\mathbf{a}}(S)^\top = \left(\sum_{i \in S} p_i(S) \mathbf{a}_i \right) \left(\sum_{i \in S} p_i(S) \mathbf{a}_i \right)^\top \leq \left(\sum_{i \in S} p_i(S) \right) \left(\sum_{i \in S} p_i(S) \mathbf{a}_i \mathbf{a}_i^\top \right) = (1 - p_0(S)) \bar{\mathbf{A}}(S).$$

Rearranging yields $\mathbf{I}(S) = \bar{\mathbf{A}}(S) - \bar{\mathbf{a}}(S) \bar{\mathbf{a}}(S)^\top \geq p_0(S) \bar{\mathbf{A}}(S)$, hence $\bar{\mathbf{A}}(S) \leq \frac{1}{p_0(S)} \mathbf{I}(S)$ and therefore

$$\bar{\mathbf{a}}(S) \bar{\mathbf{a}}(S)^\top \leq (1 - p_0(S)) \bar{\mathbf{A}}(S) \leq \frac{1 - p_0(S)}{p_0(S)} \mathbf{I}(S).$$

Since $\Delta(\hat{\pi}) = \mathbb{E}_{S \sim \hat{\pi}} [\bar{\mathbf{a}}(S) \bar{\mathbf{a}}(S)^\top] - \mathbb{E}_{S \sim \hat{\pi}} [\bar{\mathbf{a}}(S)] \left(\mathbb{E}_{S \sim \hat{\pi}} [\bar{\mathbf{a}}(S)] \right)^\top \leq \mathbb{E}[\bar{\mathbf{a}}(S) \bar{\mathbf{a}}(S)^\top]$, it follows that

$$\Delta(\hat{\pi}) \leq \mathbb{E}_{S \sim \hat{\pi}} \left[\frac{1 - p_0(S)}{p_0(S)} \mathbf{I}(S) \right].$$

Let $\rho(S) := \frac{1 - p_0(S)}{p_0(S)}$. Then

$$\Delta(\hat{\pi}) \leq \mathbb{E}[\rho(S) \mathbf{I}(S)] \leq \left(\sup_{S \sim \hat{\pi}} \rho(S) \right) \mathbb{E}[\mathbf{I}(S)] = \left(\sup_{S \sim \hat{\pi}} \rho(S) \right) \mathbf{M}(\hat{\pi}),$$

which implies $\epsilon_{\text{lift}} \leq \sup_{S \sim \hat{\pi}} \rho(S)$ by the definition of ϵ_{lift} . The specialization follows from $\rho(S) = \sum_{j \in S} \exp(\mathbf{a}_j^\top \theta_0) \leq |S| e^B \leq K e^B$. Combining this with Proposition B.4, we conclude the proof. $\square$

C. Supplementary Material for Section 4

C.1. Proof of Theorem 4.4

We begin by stating an extended version of Theorem 4.4, which additionally yields a potentially sharper bound in the uniform revenue setting ($r_i \equiv 1$), particularly when the assortment size K is large. In what follows, we prove Theorem C.1 rather than Theorem 4.4, since the former is more comprehensive and subsumes the latter.

Theorem C.1 (Sample complexity, extended version of Theorem 4.4). *Let $\delta \in (0, 1]$, $T_0 = \tilde{\Omega}(B^2 \kappa^{-1} d^2 \log(N/\delta))$, and $\lambda = 1$. Suppose that Assumption 4.1 holds. Using the stopping criterion in (C.3), with probability at least $1 - \delta$, Algorithm 2, when implemented with the MILP solver described in Subsection 3.4, returns the optimal assortment S^* , and the total number of samples collected up to the stopping time τ satisfies*

$$\tau = \begin{cases} \tilde{O}\left(d \log(N/\delta) \left(\frac{1}{\Delta_{\min}^2} + \frac{1}{\kappa \Delta_{\min}} \right) + T_0\right) & \text{(Non-uniform } r_i, \text{ MILP solver),} \\ \tilde{O}\left(d \log(N/\delta) \left(\min \left\{ 1, \frac{e^B}{K} \right\} \frac{1}{\Delta_{\min}^2} + \frac{1}{\kappa \Delta_{\min}} \right) + T_0\right), & \text{(Uniform } r_i, \text{ MILP solver)} \end{cases}$$

Loss and Hessian matrix. Let $\mathcal{D}_t = \{(i_s, S_s)\}_{s=1}^t$. Define $\mathcal{L}_{\mathcal{D}_t}(\boldsymbol{\theta})$ as the negative log-likelihood of $\boldsymbol{\theta}$ with respect to data collected up to t , and the corresponding maximum likelihood estimate (MLE):

$$\mathcal{L}_{\mathcal{D}_t}(\boldsymbol{\theta}) := \sum_{s=1}^t \ell_s(\boldsymbol{\theta}) + \frac{\lambda}{2} \|\boldsymbol{\theta}\|_2^2 = - \sum_{s=1}^t \sum_{i \in S_s} y_{si} \log p(i|S_s, \boldsymbol{\theta}) + \frac{\lambda}{2} \|\boldsymbol{\theta}\|_2^2, \quad \text{and } \hat{\boldsymbol{\theta}}_t := \underset{\boldsymbol{\theta} \in \mathbb{R}^d}{\operatorname{argmin}} \mathcal{L}_{\mathcal{D}_t}(\boldsymbol{\theta}).$$

Define the regularized Hessian matrix at $\boldsymbol{\theta}$ as follows:

$$\begin{aligned} \mathbf{H}_t(\boldsymbol{\theta}) = \mathbf{H}_{\mathcal{D}_t}(\boldsymbol{\theta}) &:= \sum_{s=1}^t \nabla^2 \ell_s(\boldsymbol{\theta}) + \lambda \mathbf{I}_d = \sum_{s=1}^t \sum_{i \in S_s} p(i|S_s, \boldsymbol{\theta}) (\mathbf{a}_i - \bar{\mathbf{a}}_{\boldsymbol{\theta}}(S_s)) (\mathbf{a}_i - \bar{\mathbf{a}}_{\boldsymbol{\theta}}(S_s))^\top + \lambda \mathbf{I}_d \\ &= \sum_{s=1}^t \left(\sum_{i \in S_s} p(i|S_s, \boldsymbol{\theta}) \mathbf{a}_i \mathbf{a}_i^\top - \sum_{i \in S_s} \sum_{j \in S_s} p(i|S_s, \boldsymbol{\theta}) p(j|S_s, \boldsymbol{\theta}) \mathbf{a}_i \mathbf{a}_j^\top \right) + \lambda \mathbf{I}_d. \end{aligned}$$

Confidence bound and good event. The following proposition provides a high-probability estimation error bound for the regularized MLE and guarantees local stability of the empirical Hessian under a sufficiently small uncertainty condition.

Lemma C.2 (Extended version of Lemma 4.2, Theorem 1 of Han et al. 2026). *Given $\mathcal{D} := \{(i_t, S_t)\}_{t=1}^T$, let $\hat{\boldsymbol{\theta}}_{\mathcal{D}}$ be the (unconstrained) λ -regularized MLE under $\mathcal{D}$, i.e., $\hat{\boldsymbol{\theta}}_{\mathcal{D}} \in \underset{\boldsymbol{\theta} \in \mathbb{R}^d}{\operatorname{argmin}} \mathcal{L}_{\mathcal{D}}(\boldsymbol{\theta})$, where $\mathcal{L}_{\mathcal{D}}(\boldsymbol{\theta}) := \sum_{t=1}^T \ell_t(\boldsymbol{\theta}) + \frac{\lambda}{2} \|\boldsymbol{\theta}\|_2^2$. Assume that, conditional on $\{S_t\}_{t \in [T]}$, the observed choices $\{i_t\}_{t \in [T]}$ are mutually independent and that Assumption 4.1 holds. Define $\mathbf{H}_{\mathcal{D}}(\boldsymbol{\theta}) := \sum_{t \in [T]} \nabla^2 \ell_t(\boldsymbol{\theta}) + \lambda \mathbf{I}_d$. Suppose that the event*

$$64 \max_{s \leq t, i \in S_s} \|\mathbf{a}_i\|_{\mathbf{H}_{\mathcal{D}}(\boldsymbol{\theta}^*)^{-1}} \leq \frac{1}{\sqrt{d \log(N/\delta)}} \wedge \frac{1}{\sqrt{\lambda} B}$$

holds. Then, with probability at least $1 - \delta$, we have

$$(i) \quad \frac{1}{2} \mathbf{H}_{\mathcal{D}}(\boldsymbol{\theta}^*) \leq \mathbf{H}_{\mathcal{D}}(\hat{\boldsymbol{\theta}}_{\mathcal{D}}) \leq 2 \mathbf{H}_{\mathcal{D}}(\boldsymbol{\theta}^*).$$

$$(ii) \quad \left| \mathbf{a}_i^\top (\hat{\boldsymbol{\theta}}_{\mathcal{D}} - \boldsymbol{\theta}^*) \right| \leq \|\mathbf{a}_i\|_{\mathbf{H}_{\mathcal{D}}(\boldsymbol{\theta}^*)^{-1}} \left(36 \sqrt{\log(N/\delta)} + 64 \sqrt{\lambda} B \right), \quad \forall i \in [N].$$

Define the confidence bound

$$\beta(\delta) := 36 \sqrt{\log \frac{N}{\delta}} + 64 \sqrt{\lambda} B. \quad (\text{C.1})$$

Let T_0 denote the number of initial warm-up agent–environment interactions. For each $t > T_0$, we define a *good event*

$$\mathcal{E} := \bigcap_{t > T_0} \mathcal{E}_t, \quad \text{where } \mathcal{E}_t := \left\{ \forall i \in [N], \left| \mathbf{a}_i^\top (\hat{\boldsymbol{\theta}}_t - \boldsymbol{\theta}^*) \right| \leq \beta(\delta) \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}^*)^{-1}} \right\}, \quad (\text{C.2})$$

Lemma C.3 (Bounded length of the exploration phase, Lemma 1 of Han et al. 2026). *In Algorithm 2, there exists an absolute constant $c_0 > 0$ such that the exploration phase terminates after at most $c_0 \kappa^{-1} d (\sqrt{d \log(N/\delta)} \vee B \sqrt{\lambda})^2 \log(dN B \lambda / \delta)$ iterations. Moreover, it holds that*

$$256 \max_{i \in [N]} \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}^*)^{-1}} \leq \frac{1}{\sqrt{d \log(N/\delta)}} + \frac{1}{B \sqrt{\lambda}}.$$

Lemma C.3 guarantees that choosing $T_0 = \tilde{\Omega} \left(\kappa^{-1} d (\sqrt{d \log(N/\delta)} \vee B \sqrt{\lambda})^2 \right)$ ensures that, for all $t > T_0$, the conditions of Lemma C.2 hold; hence Lemma C.2 applies and the event in (C.2) occurs.

Optimal design. After the warm-up rounds, we compute the warm-up parameter as

$$\boldsymbol{\theta}_0 \leftarrow \underset{\boldsymbol{\theta} \in \mathbb{R}^d}{\operatorname{argmin}} \mathcal{L}_{\mathcal{D}'_w}(\boldsymbol{\theta}),$$

where $\mathcal{D}'_w$ contains the same assortments as $\mathcal{D}_w$ (which will be continually updated during the main rounds), but with independently generated choice feedback for each assortment.

Then, given the warm-up parameter $\boldsymbol{\theta}_0$, we compute the optimal design $\hat{\pi}_{\boldsymbol{\theta}_0}$ by running the Frank–Wolfe algorithm (Algorithm 1). Note that this design can be computed either using a 0–1 *MILP solver* (as in Subsection 3.4) or via a relaxed but polynomial-time *lifted G-optimal design* formulation (as in Subsection 3.5).

Optimistic and pessimistic revenue. Recall the warm-up parameter $\boldsymbol{\theta}_0 \leftarrow \underset{\boldsymbol{\theta} \in \mathbb{R}^d}{\operatorname{argmin}} \mathcal{L}_{\mathcal{D}'_w}(\boldsymbol{\theta})$. Then, we define the optimistic/pessimistic utilities at time t as:

$$u_{t,i}^+ = \mathbf{a}_i^\top \hat{\boldsymbol{\theta}}_t + \sqrt{2}\beta(\delta) \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}, \quad u_{t,i}^- = \mathbf{a}_i^\top \hat{\boldsymbol{\theta}}_t - \sqrt{2}\beta(\delta) \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}.$$

Moreover, for any $S \in \mathcal{S}$, define the optimistic/pessimistic expected revenues at time t as:

$$\tilde{R}_t(S) := \frac{\sum_{i \in S} \exp(u_{t,i}^+) r_i}{1 + \sum_{j \in S} \exp(u_{t,j}^+)}, \quad \check{R}_t(S) := \frac{\sum_{i \in S} \exp(u_{t,i}^-) r_i}{1 + \sum_{j \in S} \exp(u_{t,j}^-)}.$$

At each round t , we select the pessimistic best assortment and the optimistic best alternative as

$$S_t^{\text{best}} \in \underset{S \in \mathcal{S}}{\operatorname{argmax}} \check{R}_t(S), \quad S_t^{\text{alt}} \in \underset{S \in \mathcal{S}: S \neq S_t^{\text{best}}}{\operatorname{argmax}} \tilde{R}_t(S).$$

Lemma C.4 (Optimistic/pessimistic revenue gaps S_t^{best} and S_t^{alt}). *For any $t > T_0$, let $S_t^{\text{best}} \in \underset{S \in \mathcal{S}}{\operatorname{argmax}} \check{R}_t(S)$ and $S_t^{\text{alt}} \in \underset{S \in \mathcal{S}, S \neq S_t^{\text{best}}}{\operatorname{argmax}} \tilde{R}_t(S)$. On the event $\mathcal{E}$ in (C.2), we have*

$$|R(S_t^{\text{best}}, \boldsymbol{\theta}^*) - \check{R}_t(S_t^{\text{best}})| \leq 2\sqrt{2}e^{9/4}\beta(\delta) \sqrt{\operatorname{Var}(r_i | S_t^{\text{best}}, \boldsymbol{\theta}^*)} \sqrt{\max_{S \in \mathcal{S}} \operatorname{tr}(\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1} \mathbf{I}_{\boldsymbol{\theta}_0}(S))} + 20\beta(\delta)^2 \max_{i \in S_t^{\text{best}}} \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}^2,$$

and

$$|R(S_t^{\text{alt}}, \boldsymbol{\theta}^*) - \tilde{R}_t(S_t^{\text{alt}})| \leq 2\sqrt{2}e^{9/4}\beta(\delta) \sqrt{\operatorname{Var}(r_i | S_t^{\text{alt}}, \boldsymbol{\theta}^*)} \sqrt{\max_{S \in \mathcal{S}} \operatorname{tr}(\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1} \mathbf{I}_{\boldsymbol{\theta}_0}(S))} + 20\beta(\delta)^2 \max_{i \in S_t^{\text{alt}}} \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}^2.$$

The proof is deferred to Appendix C.3.1.

Lemma C.5 (Variance bound for revenue parameters). *For any $S \in \mathcal{S}$, the following bounds hold:*

- (i) (**Non-uniform revenues**) *For arbitrary revenue parameters $(r_i)_{i \in [N]}$, we have $\operatorname{Var}(r_i | S, \boldsymbol{\theta}^*) \leq R(S^*, \boldsymbol{\theta}^*) \leq 1$.*
- (ii) (**Uniform revenues**) *If $r_i = 1$ for all $i \in [N]$, then $\operatorname{Var}(r_i | S, \boldsymbol{\theta}^*) \leq \sum_{i \in S} p(i | S, \boldsymbol{\theta}^*) p(0 | S, \boldsymbol{\theta}^*) \leq \min \left\{ 1, \frac{e^B}{|S|} \right\}$.*

The proof is deferred to Appendix C.3.2.

Lemma C.6. *Set $\lambda = 1$. Assume that after running Algorithm 1, the resulting design satisfies $g_{\boldsymbol{\theta}_0}(\hat{\pi}_{\boldsymbol{\theta}_0}) \leq (1 + \epsilon)d$, where $\epsilon > 1$. Under the event $\mathcal{E}$, for any $t > T_0$, with probability at least $1 - \delta$, the following holds for all $t > T_0$:*

- (i) $\max_{S \in \mathcal{S}} \operatorname{tr}(\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1} \mathbf{I}_{\boldsymbol{\theta}_0}(S)) \leq \frac{96(1 + \epsilon)d \log(2t)}{t - T_0}.$
- (ii) $\max_{i \in [N]} \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}^2 \leq \frac{96e(1 + \epsilon)d \log(2t)}{\kappa(t - T_0)}.$

The proof is deferred to Appendix C.3.3.

Stopping rule. We stop at the first time τ such that

$$\check{R}_\tau(S_\tau^{\text{best}}) > \check{R}_\tau(S_\tau^{\text{alt}}), \quad (\text{C.3})$$

and output S_τ^{best} . Then, the following lemma shows that $S_\tau^{\text{best}} = S^*$.

Lemma C.7. *On the event $\mathcal{E}$, if $\check{R}_\tau(S_\tau^{\text{best}}) > \check{R}_\tau(S_\tau^{\text{alt}})$, then the selected assortment at τ is optimal, i.e., $S_\tau^{\text{best}} = S^*$.*

The proof is deferred to Appendix C.3.4.

We are now ready to provide the proof of Theorem C.1.

Proof of Theorem C.1. Throughout the proof, we condition on the event $\mathcal{E}$, which occurs with probability at least $1 - \delta$. To simplify the presentation, for any $t > T_0$, we define

$$\gamma_t(S) := 2\sqrt{2}e^{9/4}\beta(\delta)\sqrt{\text{Var}(r_i|S, \theta^*)}\sqrt{\max_{S \in \mathcal{S}} \text{tr}(\mathbf{H}_t(\theta_0)^{-1}\mathbf{I}_{\theta_0}(S))} + 20\beta(\delta)^2 \max_{i \in S} \|\mathbf{a}_i\|_{\mathbf{H}_t(\theta_0)^{-1}}^2$$

Let $\Delta_t := \min_{S \neq S_t^{\text{best}}} (R(S_t^{\text{best}}, \theta^*) - R(S, \theta^*))$. Then, for all $S \neq S_t^{\text{best}}$, we have

$$\begin{aligned} \check{R}_t(S_t^{\text{best}}) - \check{R}_t(S) &\geq \check{R}_t(S_t^{\text{best}}) - \check{R}_t(S_t^{\text{alt}}) && (S_t^{\text{alt}} \in \arg\max_{S \in \mathcal{S}: S \neq S_t^{\text{best}}} \check{R}_t(S)) \\ &\geq R(S_t^{\text{best}}, \theta^*) - \gamma_t(S_t^{\text{best}}) - (R(S_t^{\text{alt}}, \theta^*) + \gamma_t(S_t^{\text{alt}})) && (\text{Lemma C.4}) \\ &\geq \Delta_t - \gamma_t(S_t^{\text{best}}) - \gamma_t(S_t^{\text{alt}}). \end{aligned}$$

Hence, if

$$\gamma_t(S_t^{\text{best}}) + \gamma_t(S_t^{\text{alt}}) < \Delta_t, \quad (\text{C.4})$$

then $\check{R}_t(S_t^{\text{best}}) > \check{R}_t(S)$ for all $S \neq S_t^{\text{best}}$, which is a sufficient condition for stopping:

$$\check{R}_t(S_t^{\text{best}}) > \max_{S \neq S_t^{\text{best}}} \check{R}_t(S) = \check{R}_t(S_t^{\text{alt}})$$

Therefore, the stopping inequality holds at time t .

Importantly, by Lemma C.7, once the stopping condition is satisfied, the selected assortment is optimal, i.e., $S_t^{\text{best}} = S^*$, which implies $\Delta_t = \Delta_{\min} = \min_{S \neq S^*} (R(S^*, \theta^*) - R(S, \theta^*))$ at the stopping time $t = \tau$. Therefore, in order to ensure (C.4), it suffices to upper bound

$$\gamma_\tau(S^*) + \gamma_\tau(S_\tau^{\text{alt}}),$$

where we have used the fact that $S_\tau^{\text{best}} = S^*$ at any stopping time τ .

(i) Non-uniform revenues. We first consider the general non-uniform revenue parameters.

By (C.4), any τ such that $\gamma_\tau(S^*) + \gamma_\tau(S_\tau^{\text{alt}}) < \Delta_{\min}$ forces stopping by (or at) time τ . We bound $\gamma_\tau(S^*) + \gamma_\tau(S_\tau^{\text{alt}})$ as follows:

$$\begin{aligned} \gamma_\tau(S^*) + \gamma_\tau(S_\tau^{\text{alt}}) &\leq 2 \max_{S \in \mathcal{S}} \gamma_\tau(S) \\ &= \max_{S \in \mathcal{S}} \left[4\sqrt{2}e^{9/4}\beta(\delta)\sqrt{\text{Var}(r_i|S, \theta^*)}\sqrt{\max_{S' \in \mathcal{S}} \text{tr}(\mathbf{H}_\tau(\theta_0)^{-1}\mathbf{I}_{\theta_0}(S'))} + 40\beta(\delta)^2 \max_{i \in S} \|\mathbf{a}_i\|_{\mathbf{H}_\tau(\theta_0)^{-1}}^2 \right] \\ &\leq 4\sqrt{2}e^{9/4}\beta(\delta)\sqrt{\max_{S \in \mathcal{S}} \text{tr}(\mathbf{H}_\tau(\theta_0)^{-1}\mathbf{I}_{\theta_0}(S))} + 40\beta(\delta)^2 \max_{i \in [N]} \|\mathbf{a}_i\|_{\mathbf{H}_\tau(\theta_0)^{-1}}^2 && (\text{Lemma C.5}) \\ &\leq 32\sqrt{3}e^{9/4}\beta(\delta)\sqrt{\frac{(1+\epsilon)d \log(2t)}{\tau - T_0}} + \beta(\delta)^2 \frac{3840e(1+\epsilon)d \log(2t)}{\kappa(\tau - T_0)}. && (\text{Lemma C.6, w.p. } 1 - \delta) \end{aligned}$$

Therefore,

$$32\sqrt{3}e^{9/4}\beta(\delta)\sqrt{\frac{(1+\epsilon)d \log(2t)}{\tau - T_0}} + \beta(\delta)^2 \frac{3840e(1+\epsilon)d \log(2t)}{\kappa(\tau - T_0)} < \Delta_{\min}. \quad (\text{C.5})$$

By the definition of $\beta(\delta)$ in (C.1), and by solving the optimal design problem using a 0–1 MILP solver (as described in Subsection 3.4), which guarantees a bounded approximation error (by Proposition 3.4), we obtain

$$\tau = \tilde{O}\left(d \log(N/\delta) \left(\frac{1}{\Delta_{\min}^2} + \frac{1}{\kappa \Delta_{\min}} \right) + T_0\right). \quad (\text{Non-uniform } r_i, \text{ MILP solver})$$

This completes the proof for the non-uniform revenue case.

(ii) Uniform revenues. We now consider the uniform revenue setting, where $r_i = 1$ for all $i \in [N]$. In this case, the expected revenue—both $\tilde{R}_t(S)$ and $\tilde{R}_t(S)$ —is monotone increasing with respect to the assortment size. Consequently, the optimal assortment, the selected assortment, and its best alternative all have maximum cardinality, i.e., $|S^*| = |S_t^{\text{best}}| = |S_t^{\text{alt}}| = K$ for all t . Hence, it suffices to bound $\gamma_\tau(S^*) + \gamma_\tau(S_t^{\text{alt}}) \leq 2 \max_{S: |S|=K} \gamma_\tau(S)$. Moreover, by Lemma C.5, we have

$$\text{Var}(r_i|S, \theta^*) \leq \min \left\{ 1, \frac{e^B}{K} \right\}.$$

This yields a potentially improved bound that explicitly captures the dependence on K . Using this variance bound and following the same proof as in the non-uniform revenue case, we obtain

$$\tau = \tilde{O}\left(d \log(N/\delta) \left(\min \left\{ 1, \frac{e^B}{K} \right\} \frac{1}{\Delta_{\min}^2} + \frac{1}{\kappa \Delta_{\min}} \right) + T_0\right). \quad (\text{Uniform } r_i, \text{ MILP solver})$$

Setting $\delta \leftarrow \delta/2$ completes the proof of Theorem C.1, and hence also of Theorem 4.4. $\square$

C.2. Proofs of Corollary 4.5

In this subsection, we provide the proof of Corollary 4.5. As in Appendix C.1, we first state an extended version of Corollary 4.5, which also encompasses the uniform revenue case.

Corollary C.8 (Sample complexity, extended version of Corollary 4.5). *Let $\delta \in (0, 1]$, $T_0 = \tilde{\Omega}(B^2 \kappa^{-1} d^2 \log(N/\delta))$, and $\lambda = 1$. Suppose that Assumption 4.1 holds. Using the stopping criterion in (C.3), with probability at least $1 - \delta$, Algorithm 2, when implemented with the lifted G-optimal design described in Subsection 3.5, returns the optimal assortment S^* , and the total number of samples collected up to the stopping time τ satisfies*

$$\tau = \begin{cases} \tilde{O}\left((1 + \epsilon_{\text{lift}})d \log(N/\delta) \left(\frac{1}{\Delta_{\min}^2} + \frac{1}{\kappa \Delta_{\min}} \right) + T_0\right) & (\text{Non-uniform } r_i, \text{ Lifted G-optimal}), \\ \tilde{O}\left((1 + \epsilon_{\text{lift}})d \log(N/\delta) \left(\min \left\{ 1, \frac{e^B}{K} \right\} \frac{1}{\Delta_{\min}^2} + \frac{1}{\kappa \Delta_{\min}} \right) + T_0\right), & (\text{Uniform } r_i, \text{ Lifted G-optimal}). \end{cases}$$

Proof of Corollary C.8. The result follows immediately by replacing the Frank–Wolfe approximation factor $(1 + \epsilon)$ in Theorem C.1 with $2(1 + \epsilon_{\text{lift}})(1 + \epsilon)$ according to Theorem 3.5. $\square$

C.3. Proofs of Lemmas for Theorem 4.4

C.3.1. PROOF OF LEMMA C.4

Proof of Lemma C.4. For any $S \in \mathcal{S}$, define $\mathcal{Q}_S : \mathbb{R}^{|S|} \rightarrow \mathbb{R}$ by

$$\mathcal{Q}_S(\mathbf{u}) := \sum_{i \in S} \frac{e^{u_i} r_i}{1 + \sum_{j \in S} e^{u_j}}, \quad \mathbf{u} = (u_i)_{i \in S} \in \mathbb{R}^{|S|}.$$

Let $\mathbf{u}^* := (u_i^*)_{i \in S} = (\mathbf{a}_i^\top \theta^*)_{i \in S}$, $\mathbf{u}_t^+ := (u_{t,i}^+)_{i \in S} \in \mathbb{R}^{|S|}$, and $\mathbf{u}_t^- := (u_{t,i}^-)_{i \in S} \in \mathbb{R}^{|S|}$. We write

$$p(i|S, \mathbf{u}) := \frac{e^{u_i}}{1 + \sum_{j \in S} e^{u_j}} \quad (i \in S), \quad p(0|S, \mathbf{u}) := \frac{1}{1 + \sum_{j \in S} e^{u_j}}.$$

Step 1: Applying the Taylor expansion. Define $\Delta u_{t,i} := u_i^* - u_{t,i}^-$ and $\Delta \mathbf{u}_t := (\Delta u_{t,i})_{i \in S_t^{\text{best}}}$, so that $\mathbf{u}^* = \mathbf{u}_t^- + \Delta \mathbf{u}_t$. Then, by the Taylor expansion, there exists $c \in (0, 1)$ and $\bar{\mathbf{u}}_t := (1 - c)\mathbf{u}_t^- + c\mathbf{u}^*$ such that

$$\begin{aligned} |R(S_t^{\text{best}}, \theta^*) - \tilde{R}_t(S_t^{\text{best}})| &= |\mathcal{Q}_{S_t^{\text{best}}}(\mathbf{u}^*) - \mathcal{Q}_{S_t^{\text{best}}}(\mathbf{u}_t^-)| \\ &\leq \left| \nabla_{\mathbf{u}} \mathcal{Q}_{S_t^{\text{best}}}(\mathbf{u}_t^-)^\top \Delta \mathbf{u}_t \right| + \frac{1}{2} \Delta \mathbf{u}_t^\top \nabla_{\mathbf{u}}^2 \mathcal{Q}_{S_t^{\text{best}}}(\bar{\mathbf{u}}_t) \Delta \mathbf{u}_t. \end{aligned} \quad (\text{C.6})$$

For the first term on the right-hand side of (C.6), we have

$$\begin{aligned}
 \left| \nabla_{\mathbf{u}} \mathcal{Q}_{S_t^{\text{best}}}(\mathbf{u}_t^-)^\top \Delta \mathbf{u}_t \right| &\leq \sum_{i \in S_t^{\text{best}}} p(i|S_t^{\text{best}}, \mathbf{u}_t^-) |r_i - \mathcal{Q}_{S_t^{\text{best}}}(\mathbf{u}_t^-)| |\Delta u_{t,i}| \\
 &= \sum_{i \in S_t^{\text{best}}} p(i|S_t^{\text{best}}, \mathbf{u}_t^-) (r_i - \check{R}_t(S_t^{\text{best}})) |\Delta u_{t,i}| \quad (r_i \geq \check{R}_t(S_t^{\text{best}}) = \mathcal{Q}_{S_t^{\text{best}}}(\mathbf{u}_t^-)) \\
 &\leq 2\sqrt{2}\beta(\delta) \sum_{i \in S_t^{\text{best}}} p(i|S_t^{\text{best}}, \mathbf{u}_t^-) (r_i - \check{R}_t(S_t^{\text{best}})) \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}, \tag{C.7}
 \end{aligned}$$

where the first equality holds by the optimality of S_t^{best} ; otherwise, if there existed $i \in S_t^{\text{best}}$ with $r_i < \check{R}_t(S_t^{\text{best}})$, removing item i would strictly increase the objective, contradicting optimality. The last inequality follows from the bound on $|\Delta u_{t,i}|$:

$$\begin{aligned}
 |\Delta u_{t,i}| &= |u_i^* - u_{t,i}^-| = \left| \mathbf{a}_i^\top (\boldsymbol{\theta}^* - \hat{\boldsymbol{\theta}}_t) + \sqrt{2}\beta(\delta) \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}} \right| \\
 &\leq \beta(\delta) \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}^*)^{-1}} + \sqrt{2}\beta(\delta) \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}} \quad (\text{Under } \mathcal{E}) \\
 &\leq 2\sqrt{2}\beta(\delta) \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}. \quad (\text{Lemma C.2, } \mathbf{H}_t(\boldsymbol{\theta}^*) \geq \frac{1}{2}\mathbf{H}_t(\boldsymbol{\theta}_0))
 \end{aligned}$$

Remark C.9 (Conditional independence via two-feedback sample splitting). Note that the conditional independence requirement in Lemma C.2 holds under our two-feedback sample-splitting scheme. In the warm-up phase, for each offered assortment S_t^{best} we collect two conditionally i.i.d. MNL choices and split them into two independent streams: one stream forms the independent-feedback warm-up dataset $\mathcal{D}'_w$, which is used to compute the warm-up estimate $\boldsymbol{\theta}_0$, while the other stream $\mathcal{D}_w$ is reserved as the main-round data used in Lemma C.2. Consequently, even if the sequence of assortments $\{S_t^{\text{best}}\}_{t \in [T]}$ is chosen adaptively based on $\mathcal{D}'_w$, the main-round observed choices $\{i_t\}_{t \in [T]}$ (i.e., the feedback contained in $\mathcal{D}_w$) are mutually independent conditional on $\{S_t^{\text{best}}\}_{t \in [T]}$, and are also independent of the randomness used to construct $\boldsymbol{\theta}_0$.

Set $r_i := 0$ and $\mathbf{a}_0 := 0$. For simplicity, we write $\mathbb{E}_{\mathbf{u}}[X_j] = \mathbb{E}_{j \sim p(\cdot|S_t^{\text{best}}, \mathbf{u})}[X_j]$ and $\text{Var}_{\mathbf{u}}[X_j] = \text{Var}_{j \sim p(\cdot|S_t^{\text{best}}, \mathbf{u})}[X_j]$ where j is a random index drawn according to the MNL choice distribution $p(\cdot|S_t^{\text{best}}, \mathbf{u})$. Then, we can further bound the right-hand side of (C.7) as:

$$\begin{aligned}
 \left| \nabla_{\mathbf{u}} \mathcal{Q}_{S_t^{\text{best}}}(\mathbf{u}_t^-)^\top \Delta \mathbf{u}_t \right| &\leq 2\sqrt{2}\beta(\delta) \sum_{i \in S_t^{\text{best}}} p(i|S_t^{\text{best}}, \mathbf{u}_t^-) (r_i - \check{R}_t(S_t^{\text{best}})) \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}} \\
 &= 2\sqrt{2}\beta(\delta) \sum_{i \in S_t^{\text{best}}} p(i|S_t^{\text{best}}, \mathbf{u}_t^-) r_i \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}} - \mathbb{E}_{\mathbf{u}_t^-}[r_j] \mathbb{E}_{\mathbf{u}_t^-}[\|\mathbf{a}_j\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}] \\
 &= 2\sqrt{2}\beta(\delta) \sum_{i \in S_t^{\text{best}} \cup \{0\}} p(i|S_t^{\text{best}}, \mathbf{u}_t^-) \left(r_i - \mathbb{E}_{\mathbf{u}_t^-}[r_j] \right) \left(\|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}} - \mathbb{E}_{\mathbf{u}_t^-}[\|\mathbf{a}_j\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}] \right) \\
 &\leq 2\sqrt{2}\beta(\delta) \sqrt{\text{Var}_{\mathbf{u}_t^-}(r_i)} \sqrt{\text{Var}_{\mathbf{u}_t^-}(\|\mathbf{a}_j\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}})}. \tag{C.8}
 \end{aligned}$$

Step 2: Ratio between $p(i|S_t^{\text{best}}, \mathbf{u}_t^-)$ and $p(i|S_t^{\text{best}}, \mathbf{u}^*)$ under $\mathcal{E}$. Under the event $\mathcal{E}$, we have

$$\begin{aligned}
 |u_{t,i}^- - u_i^*| &\leq 2\sqrt{2}\beta(\delta) \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}} \\
 &= 2\sqrt{2} \left(36\sqrt{\log(N/\delta)} + 64\sqrt{\lambda}B \right) \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}} \quad (\text{Definition of } \beta(\delta), \text{ Eqn. (C.1)}) \\
 &\leq 4 \left(36\sqrt{\log(N/\delta)} + 64\sqrt{\lambda}B \right) \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}^*)^{-1}} \quad (\text{Lemma C.2}) \\
 &\leq 1. \quad (\text{Lemma C.3})
 \end{aligned}$$

Moreover, for the outside option, since we set $\mathbf{a}_0 = 0$, we get $|u_{t,0}^- - u_0^*| = 0$. Hence, for all $i \in S \cup \{0\}$, we obtain

$$p(i|S_t^{\text{best}}, \mathbf{u}_t^-) = \frac{e^{u_{t,i}^-}}{1 + \sum_{j \in S} e^{u_{t,j}^-}} \leq e^2 \cdot \frac{e^{u_i^*}}{1 + \sum_{j \in S} e^{u_j^*}} = e^2 p(i|S_t^{\text{best}}, \mathbf{u}^*).$$

By the same argument, we also get

$$p(i|S_t^{\text{best}}, \mathbf{u}_t^-) \geq e^{-2} p(i|S_t^{\text{best}}, \mathbf{u}^*).$$

Combining the two bounds yields

$$e^{-2} \leq \frac{p(i|S_t^{\text{best}}, \mathbf{u}_t^-)}{p(i|S_t^{\text{best}}, \mathbf{u}^*)} \leq e^2. \quad (\text{C.9})$$

Step 3: Ratio between $p(i|S_t^{\text{best}}, \mathbf{u}^*)$ and $p(i|S_t^{\text{best}}, \mathbf{u}_0)$ under $\mathcal{E}$. Recall the warm-up parameter $\theta_0 \in \arg\min_{\theta \in \mathbb{R}^d} \mathcal{L}_{\mathcal{D}'_w}(\theta)$. Let $\mathbf{u}_0 := (u_i^{(0)})_{i \in S} = (\mathbf{a}_i^\top \theta_0)_{i \in S}$. Then, under the event $\mathcal{E}$, we have

$$|u_i^* - u_i^{(0)}| = |\mathbf{a}_i^\top (\theta^* - \theta_0)| \leq \|\mathbf{a}_i\|_{\mathbf{H}_t(\theta^*)^{-1}} \left(36\sqrt{\log(N/\delta)} + 64\sqrt{\lambda}B \right) \quad (\text{Lemma C.2})$$

$$\leq \frac{1}{4}. \quad (\text{Lemma C.3})$$

Using the same logic in Step 2, we obtain

$$e^{-1/2} \leq \frac{p(i|S_t^{\text{best}}, \mathbf{u}^*)}{p(i|S_t^{\text{best}}, \mathbf{u}_0)} \leq e^{1/2}. \quad (\text{C.10})$$

Step 4: Bound for the first order term. We return to (C.8) and invoke the results established in Steps 2 and 3.

$$\begin{aligned} \left| \nabla_{\mathbf{u}} \mathcal{Q}_{S_t^{\text{best}}}(\mathbf{u}_t^-)^\top \Delta \mathbf{u}_t \right| &\leq 2\sqrt{2}\beta(\delta) \sqrt{\text{Var}_{\mathbf{u}_t^-}(r_i)} \sqrt{\text{Var}_{\mathbf{u}_t^-}(\|\mathbf{a}_j\|_{\mathbf{H}_t(\theta_0)^{-1}})} \\ &\leq 2\sqrt{2}e^2\beta(\delta) \sqrt{\text{Var}_{\mathbf{u}^*}(r_i)} \sqrt{\text{Var}_{\mathbf{u}^*}(\|\mathbf{a}_i\|_{\mathbf{H}_t(\theta_0)^{-1}})} \quad (\text{Lemma C.15 and Eqn. (C.9)}) \\ &\leq 2\sqrt{2}e^{9/4}\beta(\delta) \sqrt{\text{Var}_{\mathbf{u}^*}(r_i)} \sqrt{\text{Var}_{\mathbf{u}_0}(\|\mathbf{a}_i\|_{\mathbf{H}_t(\theta_0)^{-1}})} \quad (\text{Lemma C.15 and Eqn. (C.10)}) \\ &\leq 2\sqrt{2}e^{9/4}\beta(\delta) \sqrt{\text{Var}_{\mathbf{u}^*}(r_i)} \sqrt{\mathbb{E}_{\mathbf{u}_0} \left[(\|\mathbf{a}_i\|_{\mathbf{H}_t(\theta_0)^{-1}} - \|\bar{\mathbf{a}}_{\theta_0}(S_t^{\text{best}})\|_{\mathbf{H}_t(\theta_0)^{-1}})^2 \right]}, \quad (\text{C.11}) \end{aligned}$$

where the last inequality follows from the fact that, for any $b \in \mathbb{R}$ and random variable Y ,

$$\begin{aligned} \mathbb{E}_p[(Y - b)^2] &= \mathbb{E}_p[(Y - \mathbb{E}_p[Y] + \mathbb{E}_p[Y] - b)^2] \\ &= \mathbb{E}_p[(Y - \mathbb{E}_p[Y])^2] + 2(\mathbb{E}_p[Y] - b) \mathbb{E}_p[Y - \mathbb{E}_p[Y]] + (\mathbb{E}_p[Y] - b)^2 \\ &= \text{Var}_p(Y) + (\mathbb{E}_p[Y] - b)^2 \geq \text{Var}_p(Y), \end{aligned}$$

and we set $b = \|\bar{\mathbf{a}}_{\theta_0}(S_t^{\text{best}})\|_{\mathbf{H}_t(\theta_0)^{-1}}$.

Continuing to bound the right-hand side of (C.11), we obtain

$$\begin{aligned} \left| \nabla_{\mathbf{u}} \mathcal{Q}_{S_t^{\text{best}}}(\mathbf{u}_t^-)^\top \Delta \mathbf{u}_t \right| &\leq 2\sqrt{2}e^{9/4}\beta(\delta) \sqrt{\text{Var}_{\mathbf{u}^*}(r_i)} \sqrt{\mathbb{E}_{\mathbf{u}_0} \left[(\|\mathbf{a}_i\|_{\mathbf{H}_t(\theta_0)^{-1}} - \|\bar{\mathbf{a}}_{\theta_0}(S_t^{\text{best}})\|_{\mathbf{H}_t(\theta_0)^{-1}})^2 \right]} \\ &\leq 2\sqrt{2}e^{9/4}\beta(\delta) \sqrt{\text{Var}_{\mathbf{u}^*}(r_i)} \sqrt{\mathbb{E}_{\mathbf{u}_0} \left[\|\mathbf{a}_i - \bar{\mathbf{a}}_{\theta_0}(S_t^{\text{best}})\|_{\mathbf{H}_t(\theta_0)^{-1}}^2 \right]} \quad (\text{Reverse triangle inequality}) \\ &= 2\sqrt{2}e^{9/4}\beta(\delta) \sqrt{\text{Var}_{\mathbf{u}^*}(r_i)} \sqrt{\text{tr}(\mathbf{H}_t(\theta_0)^{-1} \mathbf{I}_{\theta_0}(S_t^{\text{best}}))} \quad (\text{Definition of } \mathbf{I}_{\theta}(S_t^{\text{best}})) \\ &\leq 2\sqrt{2}e^{9/4}\beta(\delta) \sqrt{\text{Var}(r_i|S_t^{\text{best}}, \theta^*)} \sqrt{\max_{S \in \mathcal{S}} \text{tr}(\mathbf{H}_t(\theta_0)^{-1} \mathbf{I}_{\theta_0}(S))}, \quad (\text{C.12}) \end{aligned}$$

where, in the last inequality, we use the fact that $\text{Var}(r_i|S_t^{\text{best}}, \theta^*) = \text{Var}(r_i|S_t^{\text{best}}, \theta^*)$.

Step 5: Bound for the second-order term. We now bound the second term on the right-hand side of (C.6). Let

$p_i(\bar{\mathbf{u}}_t) = \frac{\exp(\bar{u}_{t,i})}{1 + \sum_{j=1}^{|S_t^{\text{best}}|} \exp(\bar{u}_{t,j})}$. Then, we have

$$\begin{aligned}
 \frac{1}{2} \Delta \mathbf{u}_t^\top \nabla_{\mathbf{u}}^2 \mathcal{Q}_{S_t^{\text{best}}}(\bar{\mathbf{u}}_t) \Delta \mathbf{u}_t &= \frac{1}{2} \sum_{i,j \in S_t^{\text{best}}} \Delta u_{t,i} \frac{\partial^2 \mathcal{Q}_{S_t^{\text{best}}}}{\partial i \partial j} \Delta u_{t,j} \leq \frac{1}{2} \sum_{i,j \in S_t^{\text{best}}} |\Delta u_{t,i}| \left| \frac{\partial^2 \mathcal{Q}_{S_t^{\text{best}}}}{\partial i \partial j} \right| |\Delta u_{t,j}| \\
 &= \frac{1}{2} \sum_{i \in S_t^{\text{best}}} \sum_{j \in S_t^{\text{best}}, j \neq i} |\Delta u_{t,i}| \left| \frac{\partial^2 \mathcal{Q}_{S_t^{\text{best}}}}{\partial i \partial j} \right| |\Delta u_{t,j}| + \frac{1}{2} \sum_{i \in S_t^{\text{best}}} |\Delta u_{t,i}| \left| \frac{\partial^2 \mathcal{Q}_{S_t^{\text{best}}}}{\partial i \partial i} \right| |\Delta u_{t,i}| \\
 &\leq \sum_{i \in S_t^{\text{best}}} \sum_{j \in S_t^{\text{best}}, j \neq i} |\Delta u_{t,i}| p_i(\bar{\mathbf{u}}_t) p_j(\bar{\mathbf{u}}_t) |\Delta u_{t,j}| + \frac{3}{2} \sum_{i \in S_t^{\text{best}}} (\Delta u_{t,i})^2 p_i(\bar{\mathbf{u}}_t) \quad (\text{Lemma C.11}) \\
 &\leq \sum_{i \in S_t^{\text{best}}} \sum_{j \in S_t^{\text{best}}} |\Delta u_{t,i}| p_i(\bar{\mathbf{u}}_t) p_j(\bar{\mathbf{u}}_t) |\Delta u_{t,j}| + \frac{3}{2} \sum_{i \in S_t^{\text{best}}} (\Delta u_{t,i})^2 p_i(\bar{\mathbf{u}}_t) \\
 &\leq \frac{1}{2} \sum_{i \in S_t^{\text{best}}} \sum_{j \in S_t^{\text{best}}} (\Delta u_{t,i})^2 p_i(\bar{\mathbf{u}}_t) p_j(\bar{\mathbf{u}}_t) + \frac{1}{2} \sum_{i \in S_t^{\text{best}}} \sum_{j \in S_t^{\text{best}}} (\Delta u_{t,j})^2 p_i(\bar{\mathbf{u}}_t) p_j(\bar{\mathbf{u}}_t) \\
 &\quad (\text{AM-GM inequality}) \\
 &\quad + \frac{3}{2} \sum_{i \in S_t^{\text{best}}} (\Delta u_{t,i})^2 p_i(\bar{\mathbf{u}}_t) \\
 &\leq \frac{5}{2} \sum_{i \in S_t^{\text{best}}} (\Delta u_{t,i})^2 p_i(\bar{\mathbf{u}}_t) \\
 &\leq 20\beta(\delta)^2 \max_{i \in S_t^{\text{best}}} \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}^2. \tag{C.13}
 \end{aligned}$$

Step 6: Combining the two bounds. Substituting (C.12) and (C.13) into (C.6), we obtain

$$|R(S_t^{\text{best}}, \boldsymbol{\theta}^*) - \tilde{R}_t(S_t^{\text{best}})| \leq 2\sqrt{2}e^{9/4}\beta(\delta)\sqrt{\text{Var}(r_i|S_t^{\text{best}}, \boldsymbol{\theta}^*)}\sqrt{\max_{S \in \mathcal{S}} \text{tr}(\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}\mathbf{I}_{\boldsymbol{\theta}_0}(S))} + 20\beta(\delta)^2 \max_{i \in S_t^{\text{best}}} \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}^2.$$

This proves the first statement of Lemma C.4.

Step 7: Second statement The proof of the second statement follows the same steps as in the first statement, with $\mathbf{u}_t^+$ replacing $\mathbf{u}_t^-$ throughout. In addition, we use the fact that $r_i \geq \tilde{R}_t(S_t^{\text{alt}})$ for all $i \in S_t^{\text{alt}}$, which holds directly by the optimality of S_t^{alt} . Hence, we get

$$|R(S_t^{\text{alt}}, \boldsymbol{\theta}^*) - \tilde{R}_t(S_t^{\text{alt}})| \leq 2\sqrt{2}e^{9/4}\beta(\delta)\sqrt{\text{Var}(r_i|S_t^{\text{alt}}, \boldsymbol{\theta}^*)}\sqrt{\max_{S \in \mathcal{S}} \text{tr}(\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}\mathbf{I}_{\boldsymbol{\theta}_0}(S))} + 20\beta(\delta)^2 \max_{i \in S_t^{\text{alt}}} \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}^2.$$

This concludes the proof of Lemma C.4. $\square$

C.3.2. PROOF OF LEMMA C.5

Proof of Lemma C.5. Fix any $S \in \mathcal{S}$. For simplicity, let $p_i = p(i|S, \boldsymbol{\theta}^*)$

For the uniform revenue parameters, i.e., $r_i = 1$ for all $i \in [N]$ (and $r_0 = 0$), we have

$$\begin{aligned}
 \text{Var}(r_i|S, \boldsymbol{\theta}^*) &= \sum_{i \in S \cup \{0\}} p_i \left(r_i - \sum_{j \in S \cup \{0\}} p_j r_j \right)^2 = \sum_{i \in S} p_i \left(1 - \sum_{j \in S} p_j \right)^2 + p_0 \left(\sum_{j \in S} p_j \right)^2 \\
 &= \sum_{i \in S} p_i p_0^2 + p_0 \left(\sum_{j \in S} p_j \right)^2 \\
 &= \sum_{i \in S} p_i p_0 \leq \frac{e^B}{|S|},
 \end{aligned}$$

which proves claim (i).

For the non-uniform revenues parameters, we have

$$\begin{aligned} \text{Var}(r_i|S, \boldsymbol{\theta}^*) &= \mathbb{E}[r_i^2|S, \boldsymbol{\theta}^*] - \mathbb{E}[r_i|S, \boldsymbol{\theta}^*]^2 \leq \mathbb{E}[r_i^2|S, \boldsymbol{\theta}^*] \leq \mathbb{E}[r_i|S, \boldsymbol{\theta}^*] = R(S, \boldsymbol{\theta}^*) \\ &\leq \max_{S' \in \mathcal{S}} R(S', \boldsymbol{\theta}^*) = R(S^*, \boldsymbol{\theta}^*), \end{aligned}$$

which proves the claim (ii). $\square$

C.3.3. PROOF OF LEMMA C.6

Proof of Lemma C.6. To prove the first claim, we apply the matrix Chernoff bound (Lemma C.14) with the choice $\gamma = \frac{1}{2}$ and $\bar{\lambda}_t = 24 \log(2d/\delta_t)$, where we set $\delta_t := \frac{6\delta}{\pi^2(t-T_0)^2}$. This yields

$$\begin{aligned} \mathbf{H}_t(\boldsymbol{\theta}_0) &\geq \sum_{s=T_0+1}^t \nabla^2 \ell_s(\boldsymbol{\theta}_0) + \lambda \mathbf{I}_d \geq \frac{1}{\bar{\lambda}_t} \left(\sum_{s=T_0+1}^t \mathbf{I}_{\boldsymbol{\theta}_0}(S_s) + \bar{\lambda}_t \mathbf{I}_d \right) \quad (\lambda = 1, \bar{\lambda}_t > 1) \\ &\geq \frac{1}{2\bar{\lambda}_t} (t - T_0) \mathbb{E}_{S \sim \hat{\pi}_{\boldsymbol{\theta}_0}} [\mathbf{I}_{\boldsymbol{\theta}_0}(S) + \bar{\lambda}_t \mathbf{I}_d] \quad (\text{Lemma C.14 with } \gamma = \frac{1}{2}) \\ &\geq \frac{1}{2\bar{\lambda}_t} (t - T_0) \mathbf{M}_{\boldsymbol{\theta}_0}(\hat{\pi}_{\boldsymbol{\theta}_0}). \quad (\mathbf{M}_{\boldsymbol{\theta}_0}(\hat{\pi}_{\boldsymbol{\theta}_0}) = \mathbb{E}_{S \sim \hat{\pi}_{\boldsymbol{\theta}_0}} [\mathbf{I}_{\boldsymbol{\theta}_0}(S)]) \end{aligned}$$

Therefore, with probability at least $1 - \delta_t$, we have

$$\begin{aligned} \max_{S \in \mathcal{S}} \text{tr}(\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1} \mathbf{I}_{\boldsymbol{\theta}_0}(S)) &\leq \frac{2\bar{\lambda}_t}{t - T_0} \max_{S \in \mathcal{S}} \text{tr}(\mathbf{M}_{\boldsymbol{\theta}_0}(\hat{\pi}_{\boldsymbol{\theta}_0})^{-1} \mathbf{I}_{\boldsymbol{\theta}_0}(S)) \\ &= \frac{2\bar{\lambda}_t}{t - T_0} g_{\boldsymbol{\theta}_0}(\hat{\pi}_{\boldsymbol{\theta}_0}) = \frac{96(1 + \epsilon)d \log(2t)}{t - T_0}, \end{aligned}$$

which proves the claim (i).

To prove the second claim, it suffices to upper bound $\max_{i \in [N]} \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}^2$ by an expression in trace form. Fix any $i \in [N]$, let $S' = \{i\}$. Then, we have

$$\begin{aligned} \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}^2 &= \frac{p(i|S', \boldsymbol{\theta}_0)p(0|S', \boldsymbol{\theta}_0)}{p(i|S', \boldsymbol{\theta}_0)p(0|S', \boldsymbol{\theta}_0)} \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}^2 \\ &\leq e \frac{p(i|S', \boldsymbol{\theta}_0)p(0|S', \boldsymbol{\theta}_0)}{p(i|S', \boldsymbol{\theta}^*)p(0|S', \boldsymbol{\theta}^*)} \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}^2 \quad (\text{Eqn. (C.10)}) \\ &\leq \frac{e}{\kappa} p(i|S', \boldsymbol{\theta}_0)p(0|S', \boldsymbol{\theta}_0) \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}^2 \quad (\text{Definition of } \kappa, \text{ Eqn. (8)}) \\ &= \frac{e}{\kappa} \text{tr}(\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1} \mathbf{I}_{\boldsymbol{\theta}_0}(S')) \\ &\leq \frac{e}{\kappa} \max_{S \in \mathcal{S}} \text{tr}(\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1} \mathbf{I}_{\boldsymbol{\theta}_0}(S)). \end{aligned}$$

Therefore, by applying claim (i) and taking the maximum over $i \in [N]$, we obtain

$$\max_{i \in [N]} \|\mathbf{a}_i\|_{\mathbf{H}_t(\boldsymbol{\theta}_0)^{-1}}^2 \leq \frac{96e(1 + \epsilon)d \log(2t)}{\kappa(t - T_0)}.$$

Finally, since $\sum_{t=T_0+1}^{\infty} \delta_t = \frac{6\delta}{\pi^2} \sum_{t=1}^{\infty} \frac{1}{t^2} = \delta$, a union bound over all $t > T_0$ shows that (i) and (ii) hold simultaneously for all $t > T_0$ with probability at least $1 - \delta$. This completes the proof. $\square$

C.3.4. PROOF OF LEMMA C.7

Proof of Lemma C.7. We prove the lemma by contradiction. Fix any stopping time τ and suppose that $S_\tau^{\text{best}} \neq S^*$. On the event $\mathcal{E}$, since $R(S^*, \theta^*) \leq \tilde{R}_\tau(S^*)$ by Lemma C.10, we have

$$\begin{aligned} R(S^*, \theta^*) &\leq \tilde{R}_\tau(S^*) \leq \max_{S \in \mathcal{S}: S \neq S_\tau^{\text{best}}} \tilde{R}_\tau(S) = \tilde{R}_\tau(S_\tau^{\text{alt}}) \\ &< \tilde{R}_\tau(S_\tau^{\text{best}}), \\ &\leq R(S_\tau^{\text{best}}, \theta^*). \end{aligned} \tag{Eqn. (C.3)}$$

The last inequality holds because, on the event $\mathcal{E}$, the true utilities satisfy $u_{\tau,i}^- \leq \mathbf{a}_i^\top \theta^*$ for all i , and the expected revenue is monotone in the utilities at the maximizer (see, e.g., Lemma A.3 of Agrawal et al. 2019).

This yields

$$R(S^*, \theta^*) < R(S_\tau^{\text{best}}, \theta^*),$$

which contradicts the optimality of S^* . Therefore, we must have $S_\tau^{\text{best}} = S^*$. $\square$

C.4. Technical Lemmas

Lemma C.10 (Lemma 4 of Oh & Iyengar 2021). *Let $\tilde{R}_t(S) = \frac{\sum_{i \in S} \exp(u_{t,i}^+) r_i}{1 + \sum_{j \in S} \exp(u_{t,j}^+)}$. If for every item $i \in S^*$, $u_{t,i}^+ \geq a_i^\top \theta^*$, then for all $t \geq 1$, the following inequalities hold:*

$$R(S^*, \theta^*) \leq \tilde{R}_t(S^*).$$

Lemma C.11 (Lemma E.3 of Lee & Oh 2024). *Define $\mathcal{Q} : \mathbb{R}^K \rightarrow \mathbb{R}$, such that for any $\mathbf{u} = (u_1, \dots, u_K) \in \mathbb{R}^K$, $\mathcal{Q}(\mathbf{u}) = \sum_{i=1}^K \frac{\exp(u_i)}{1 + \sum_{k=1}^K \exp(u_k)}$. Let $p_i(\mathbf{u}) = \frac{\exp(u_i)}{1 + \sum_{k=1}^K \exp(u_k)}$. Then, for all $i \in [K]$, we have*

$$\left| \frac{\partial^2 \mathcal{Q}}{\partial i \partial j} \right| \leq \begin{cases} 3p_i(\mathbf{u}) & \text{if } i = j, \\ 2p_i(\mathbf{u})p_j(\mathbf{u}) & \text{if } i \neq j. \end{cases}$$

Lemma C.12 (Proposition B.5 of Lee & Oh 2025). *For any $t \in [T]$, the Hessian of the multinomial logistic loss $\ell_t : \mathbb{R}^d \rightarrow \mathbb{R}$ satisfies the following for any $\theta_1, \theta_2 \in \mathbb{R}^d$:*

$$e^{-3\sqrt{2} \max_{i \in S_t} \|\mathbf{a}_i^\top (\theta_1 - \theta_2)\|_\infty} \nabla^2 \ell_t(\theta_1) \leq \nabla^2 \ell_t(\theta_2) \leq e^{3\sqrt{2} \max_{i \in S_t} \|\mathbf{a}_i^\top (\theta_1 - \theta_2)\|_\infty} \nabla^2 \ell_t(\theta_1).$$

Lemma C.13 (Proposition B.6 of Lee & Oh 2025). *For any $t \in [T]$, the Hessian of the multinomial logistic loss $\ell_t : \mathbb{R}^d \rightarrow \mathbb{R}$ satisfies the following for any $\mathbf{u}, \theta_1, \theta_2 \in \mathbb{R}^d$:*

$$\mathbf{u}^\top \left(\int_0^1 (1-s) \nabla^2 \ell_t(\theta_1 + s(\theta_2 - \theta_1)) ds \right) \mathbf{u} \geq \frac{1}{2 + 3\sqrt{2} \max_{i \in S_t} \|\mathbf{a}_i^\top (\theta_1 - \theta_2)\|_\infty} \mathbf{u}^\top \nabla^2 \ell_t(\theta_1) \mathbf{u}.$$

Lemma C.14 (Matrix Chernoff bound for independent PSD matrices). *Let $\{X_s\}_{s=1}^t$ be independent random matrices in $\mathbb{R}^{d \times d}$ such that $X_s \geq \mathbf{0}$ and $\|X_s\| \leq L$ almost surely for some constant $L > 0$. Define $\hat{\mathbf{H}}_t := \sum_{s=1}^t X_s$ and $\mathbf{H}_t := \mathbb{E}[\hat{\mathbf{H}}_t]$. Fix any $\gamma \in (0, 1)$ and $\delta \in (0, 1)$. Assume that $t \geq \frac{6}{\gamma^2} \log \frac{2d}{\delta}$ and choose $\bar{\lambda} = \frac{6L}{\gamma^2} \log \frac{2d}{\delta}$. Then, with probability at least $1 - \delta$, we have*

$$(1 - \gamma)(\mathbf{H}_t + \bar{\lambda} \mathbf{I}_d) \leq \hat{\mathbf{H}}_t + \bar{\lambda} \mathbf{I}_d \leq (1 + \gamma)(\mathbf{H}_t + \bar{\lambda} \mathbf{I}_d).$$

Proof of Lemma C.14. We prove the lemma by reducing the problem to a standard matrix Chernoff bound for sums of independent positive semidefinite (PSD) matrices with eigenvalues in $[0, 1]$.

Define

$$X_s^\lambda := X_s + \frac{\bar{\lambda}}{t} \mathbf{I}_d, \quad Y := \sum_{s=1}^t X_s^\lambda = \hat{\mathbf{H}}_t + \bar{\lambda} \mathbf{I}_d, \quad \mu := \mathbb{E}[Y] = \mathbf{H}_t + \bar{\lambda} \mathbf{I}_d.$$

Then $X_s^\lambda \geq \mathbf{0}$ and $\mu \geq \bar{\lambda} \mathbf{I}_d$, so $\bar{\lambda}_{\min}(\mu) \geq \bar{\lambda}$. Moreover,

$$\|X_s^\lambda\| \leq \|X_s\| + \bar{\lambda}/t \leq L + \bar{\lambda}/t \quad \text{a.s.}$$

Let $R_{\max} := \max_{s \in [t]} \|X_s^\lambda\| \leq L + \bar{\lambda}/t$.

Since $\mu \geq \bar{\lambda} \mathbf{I}_d$, we have $\mathbf{I}_d \leq \mu/\bar{\lambda}$. Hence, for every s ,

$$\mathbf{0} \leq X_s^\lambda \leq \|X_s^\lambda\| \mathbf{I}_d \leq R_{\max} \mathbf{I}_d \leq \frac{R_{\max}}{\bar{\lambda}} \mu.$$

Define the normalized matrices $Z_s := \frac{\bar{\lambda}}{R_{\max}} \mu^{-1/2} X_s^\lambda \mu^{-1/2}$. Then $Z_s \geq \mathbf{0}$ and, using $X_s^\lambda \leq (R_{\max}/\bar{\lambda})\mu$,

$$Z_s \leq \frac{\bar{\lambda}}{R_{\max}} \mu^{-1/2} \left(\frac{R_{\max}}{\bar{\lambda}} \mu \right) \mu^{-1/2} = \mathbf{I}_d,$$

so $0 \leq Z_s \leq \mathbf{I}_d$. Furthermore,

$$\mathbb{E} \left[\sum_{s=1}^t Z_s \right] = \frac{\bar{\lambda}}{R_{\max}} \mu^{-1/2} \mathbb{E} \left[\sum_{s=1}^t X_s^\lambda \right] \mu^{-1/2} = \frac{\bar{\lambda}}{R_{\max}} \mathbf{I}_d.$$

This implies

$$\lambda_{\min} \left(\mathbb{E} \left[\sum_{s=1}^t Z_s \right] \right) = \lambda_{\max} \left(\mathbb{E} \left[\sum_{s=1}^t Z_s \right] \right) = \frac{\bar{\lambda}}{R_{\max}}.$$

Applying the standard matrix Chernoff bound to the independent matrices $\{Z_s\}$ with $0 \leq Z_s \leq \mathbf{I}_d$, for any $\gamma \in (0, 1)$ we have

$$\mathbb{P} \left(\lambda_{\min} \left(\sum_{s=1}^t Z_s \right) \leq (1 - \gamma) \frac{\bar{\lambda}}{R_{\max}} \right) \leq d \cdot \exp \left(-\frac{\gamma^2}{2} \cdot \frac{\bar{\lambda}}{R_{\max}} \right), \quad (\text{C.14})$$

$$\mathbb{P} \left(\lambda_{\max} \left(\sum_{s=1}^t Z_s \right) \geq (1 + \gamma) \frac{\bar{\lambda}}{R_{\max}} \right) \leq d \cdot \exp \left(-\frac{\gamma^2}{3} \cdot \frac{\bar{\lambda}}{R_{\max}} \right). \quad (\text{C.15})$$

Therefore, if

$$\frac{\bar{\lambda}}{R_{\max}} \geq \frac{3}{\gamma^2} \log \frac{2d}{\delta}, \quad (\text{C.16})$$

then each of (C.14)–(C.15) is at most $\delta/2$, and by a union bound we obtain, with probability at least $1 - \delta$,

$$(1 - \gamma) \frac{\bar{\lambda}}{R_{\max}} \mathbf{I}_d \leq \sum_{s=1}^t Z_s \leq (1 + \gamma) \frac{\bar{\lambda}}{R_{\max}} \mathbf{I}_d. \quad (\text{C.17})$$

Multiply (C.17) on both sides by $(R_{\max}/\bar{\lambda})\mu^{1/2}(\cdot)\mu^{1/2}$ gives

$$(1 - \gamma)\mu \leq \sum_{s=1}^t X_s^\lambda \leq (1 + \gamma)\mu.$$

Recalling $\sum_{s=1}^t X_s^\lambda = \hat{\mathbf{H}}_t + \bar{\lambda} \mathbf{I}_d$ and $\mu = \mathbf{H}_t + \bar{\lambda} \mathbf{I}_d$ proves the statement provided (C.16) holds.

We now derive a sufficient condition on t under which (C.16) holds. Since $R_{\max} \leq L + \bar{\lambda}/t$, it is sufficient to guarantee that

$$\frac{\bar{\lambda}}{R_{\max}} = \frac{\bar{\lambda}}{L + \bar{\lambda}/t} \geq \frac{3}{\gamma^2} \log \frac{2d}{\delta}.$$

Choosing $\bar{\lambda} = \frac{6L}{\gamma^2} \log \frac{2d}{\delta}$ and assuming $t \geq \frac{6}{\gamma^2} \log \frac{2d}{\delta}$, we obtain

$$L + \frac{\bar{\lambda}}{t} = L + \frac{\frac{6L}{\gamma^2} \log \frac{2d}{\delta}}{t} \leq L + \frac{\frac{6L}{\gamma^2} \log \frac{2d}{\delta}}{\frac{6}{\gamma^2} \log \frac{2d}{\delta}} = 2L.$$

Therefore,

$$\frac{\bar{\lambda}}{R_{\max}} \geq \frac{\frac{6L}{\gamma^2} \log \frac{2d}{\delta}}{2L} = \frac{3}{\gamma^2} \log \frac{2d}{\delta},$$

which establishes the desired condition. $\square$

Lemma C.15 (Variance comparison under bounded likelihood ratios). *Let $p = (p_i)_{i \in \Omega}$ and $q = (q_i)_{i \in \Omega}$ be probability mass functions on a finite set Ω , and assume $q_i > 0$ for all i . Fix $c \geq 1$ and suppose that*

$$\frac{1}{c} \leq \frac{p_i}{q_i} \leq c \quad \text{for all } i \in \Omega.$$

Then for any real-valued random variable $X : \Omega \rightarrow \mathbb{R}$,

$$\text{Var}_p(X) \leq c \text{Var}_q(X).$$

Proof of Lemma C.15. Write $\eta_i := \frac{p_i}{q_i}$, so that $\eta_i \in [1/c, c]$ and $p_i = q_i \eta_i$. Recall the variational characterization of the variance: for any distribution ν on Ω ,

$$\text{Var}_\nu(X) = \min_{a \in \mathbb{R}} \mathbb{E}_\nu[(X - a)^2]. \quad (\text{C.18})$$

(Indeed, $\mathbb{E}_\nu[(X - a)^2] = \text{Var}_\nu(X) + (a - \mathbb{E}_\nu[X])^2$, minimized at $a = \mathbb{E}_\nu[X]$.)

Fix any $a \in \mathbb{R}$. Using $p_i = q_i \eta_i$ and $\eta_i \leq c$,

$$\begin{aligned} \mathbb{E}_p[(X - a)^2] &= \sum_{i \in \Omega} p_i (X_i - a)^2 = \sum_{i \in \Omega} q_i \eta_i (X_i - a)^2 \leq c \sum_{i \in \Omega} q_i (X_i - a)^2 \\ &= c \mathbb{E}_q[(X - a)^2]. \end{aligned}$$

Taking $\min_{a \in \mathbb{R}}$ on both sides and using (C.18) yields

$$\text{Var}_p(X) = \min_a \mathbb{E}_p[(X - a)^2] \leq c \min_a \mathbb{E}_q[(X - a)^2] = c \text{Var}_q(X).$$

This completes the proof. $\square$